What the visit counts remember

Which Markov chains can be told apart by their occupation times, and how well

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1 The question and the answer

A chain on d states switches slowly. We do not see its path. We only see how long it spent in each state (the count vector of Paper C, divided by N). Which rate matrices can be told apart from this, and how much is lost compared with seeing the path?

Here is what we find. The proofs are in the sections that follow.

1. *One horizon is every horizon* (Theorem 3.1). For an irreducible chain in continuous time, the law of the occupation times at a single time fixes their law at every time. (For a chain in discrete time this is open, and the first $2d$ horizons are enough, Proposition 10.1.)

2. *The counts know each pair of flows, but not which is which* (Theorem 4.1). Suppose the start law gives positive mass to every state. For two states i and j, the counts then fix the flow from i to j and the flow from j to i as an unordered pair. For a fixed start law the exit rates decide the order for almost every rate matrix, so almost every chain can be recovered (Corollary 4.3). A stationary start is the exception (item 3).

3. *A stationary start hides the arrow of time, and for a chain with no cut it hides nothing else* (Proposition 2.1, Theorem 5.2). A chain and its reversal give the same counts. These are the only two candidates whenever the rate matrix has no cut (no way to split the states into two groups of two or more that talk to each other through rank one blocks), and almost every chain is like this. The size of the net flow on every pair is still known, so the counts say how far the chain is from reversible but not which way it turns.

4. *The counts measure the entropy production* (Theorem 6.1). With a stationary start they cannot tell a film of the chain from the same film run backward. Yet they determine exactly how different the two films are, namely the entropy production rate, and from any start they decide whether the chain is reversible (when every rate is positive).

5. *Lumping gives more ties* (Theorem 7.1). Suppose a group of states looks like a single state from outside, and every entrance into the group and the start inside it follow one law. If the start of the lumped chain is stationary, we can reverse the outside and keep the inside. If the entrance law is stationary for the inside chain, we can reverse the inside and keep the outside. This gives two chains on four states, with every rate positive and a start that is not stationary, that no amount of count data can tell apart.

6. *The net flow is learned slowly, the entropy production is not* (Theorem 9.1). Near a reversible chain with a stationary start, no estimator based on n copies of the counts finds the size of

the net flow to better than $n^{-1/4}$, where the path is expected to give $n^{-1/2}$. The counts see the net flow only through its square, and at a reversible chain they have positive information about the square (Theorem 9.2). This gives a one-sided test of reversibility that sees a net flow of size $n^{-1/4}$ when the rest of the chain is known (Theorem 9.4). Near a reversible chain the entropy production is learned to within $n^{-1/2}$, also where it is itself much larger than that (Corollary 9.5). Here too the stationary law, the two-way flows and the direction of the net flow are taken as known.

7. *What is lost* (Theorems 8.1 and 8.2). When switches are rare and the start law is known, the counts are as good as the full path for the two-way flow on each pair, and blind to first order to every net flow. With a stationary start the information about the net flow vanishes at every reversible chain, at every horizon.

2 Setting

This section fixes the notation, namely the rate matrix, the occupation times, the numbers attached to each pair of states, and the transform through which we read the law.

Let $G = (g_{ij})$ be a rate matrix on $[d] = \{1, \dots, d\}$, so $g_{ij} \geq 0$ for $i \neq j$ and $q_i = -g_{ii} = \sum_{j \neq i} g_{ij}$ is the exit rate of i. Let Y be the chain with rates G and start law μ, and let

$$L_i(t) = \int_0^t \mathbf{1}\{Y_s = i\}\, ds, \qquad L(t) = (L_1(t), \dots, L_d(t)).$$

This is the limit of Paper C's setting. If X is the N-step chain with transition matrix $I + G/N$, then its count vector K_N satisfies $K_N/N \to L(1)$ in law. With $G = TQ$ the number T counts the switches, as in Papers A to C.

We use three numbers for each pair of states. The *flows* are $f_{ij} = \mu_i g_{ij}$ (how much mass moves from i to j at time 0). The *two-way flow* is $c_{ij} = f_{ij} + f_{ji}$ and the *net flow* is $J_{ij} = f_{ij} - f_{ji}$. We also write $a_{ij} = g_{ij} g_{ji}$, the weight of a round trip. Note that

$$f_{ij} + f_{ji} = c_{ij}, \qquad f_{ij} f_{ji} = \mu_i \mu_j a_{ij}, \qquad J_{ij}^2 = c_{ij}^2 - 4\mu_i \mu_j a_{ij}, \tag{1}$$

and that $\sum_j J_{ij} = -(\mu G)_i$. So μ is stationary exactly when the net flow obeys Kirchhoff's law at every state, and the chain is reversible with respect to μ exactly when $J = 0$. (In the physics literature c_{ij} is the traffic or dynamical activity of the pair and J_{ij} is its current, and the times L are the standard example of an observable that does not change when time is reversed [18].)

Write $\Lambda = \mathrm{diag}(\lambda)$. By the Feynman–Kac formula, for $\lambda \in \mathbb{C}^d$,

$$\Phi_t(\lambda) = \mathbb{E}\, e^{-\lambda \cdot L(t)} = \mu^\top e^{t(G - \Lambda)} \mathbf{1}, \qquad r(\lambda) = \int_0^\infty \Phi_t(\lambda)\, dt = \mu^\top (\Lambda - G)^{-1} \mathbf{1}, \tag{2}$$

the second for λ with positive coordinates. The function r is rational, which is what makes it easy to read.

Proposition 2.1 (reversal)**.** *Let $\mu = \pi$ be stationary for G with $\pi_i > 0$, and let $\hat{g}_{ij} = \pi_j g_{ji}/\pi_i$. Then $(\pi, \hat{G})$ and (π, G) give the same law of $L(t)$ for every t.*

Proof. Run the film backward. The path $s \mapsto Y_{t-s}$ is the stationary chain with rates $\hat{G}$, and it spends the same time in each state as Y does. □

In terms of flows, reversal swaps f_{ij} and f_{ji} on every pair. So it keeps c and flips the sign of J.

3 One horizon is every horizon

This section shows that the law of the occupation times at one time determines their law at every time.

Theorem 3.1. *Let G be irreducible. Suppose a second pair (μ', G') gives the same law of $L(1)$. Then it gives the same law of $L(t)$ for every $t > 0$, and $r' = r$.*

The idea is that each state contributes one mode $e^{z_i(\lambda)}$ to Φ_1, where z_i is an eigenvalue of $G - \Lambda$. Exponentials of different algebraic functions cannot cancel each other, so each mode can be read off from Φ_1. The modes are the poles of the resolvent, and the resolvent knows every horizon. We need two lemmas. Both are known. The first is a weak case of theorems on the independence of exponentials [4], and the second is Lemma 5.1 of [3]. We include short proofs.

Lemma 3.2 (exponentials do not cancel)**.** *Let $B \subset \mathbb{C}^d$ be a ball and let $w_1, \dots, w_m$ and $c_1, \dots, c_m$ be holomorphic functions on B that are algebraic over $\mathbb{C}(\lambda)$. Suppose $w_a - w_b$ is not constant for $a \neq b$. If $\sum_a c_a e^{w_a} = 0$ on B, then every c_a is zero.*

Proof. We use induction on m. The case $m = 1$ is clear. Suppose the claim holds for $m-1$ and that $c_m \neq 0$ (otherwise we have fewer terms). On a smaller ball where c_m has no zero, divide by $c_m e^{w_m}$ and differentiate in λ_j. With $b_a = c_a/c_m$ and $v_a = w_a - w_m$ this gives $\sum_{a<m}(\partial_j b_a + b_a \partial_j v_a)e^{v_a} = 0$, which has $m - 1$ terms. By induction $\partial_j(b_a e^{v_a}) = 0$ for every j and every $a < m$, so $b_a e^{v_a}$ is a constant k_a. If some $k_a \neq 0$, then $e^{v_a} = k_a/b_a$ is algebraic. This is impossible. Restrict to a complex line on which v_a is not constant. There v_a is a nonconstant meromorphic function on a compact Riemann surface, so it has a pole, and near the pole e^{v_a} has an essential singularity, which an algebraic function does not have. Hence every $k_a = 0$, so $c_a = 0$ for $a < m$ (on the smaller ball, and so on B), and then $c_m = 0$, which contradicts our assumption. □

Lemma 3.3 (one polynomial)**.** *If G is irreducible, then $\chi(z, \lambda) = \det(z + \Lambda - G)$ is an irreducible polynomial.*

Proof. We argue by contradiction. A factorization of χ would force a product rule on its coefficients, and a cycle of the chain breaks that rule.

Put $x_i = z + \lambda_i + q_i$, let $X = \operatorname{diag}(x_1, \dots, x_d)$, and let $A \geq 0$ be G without its diagonal. Then

$$\chi = \det(X - A) = \sum_{U \subseteq [d]} m(U) \prod_{i \notin U} x_i, \qquad m(U) = \det(-A_{UU}), \quad m(\emptyset) = 1.$$

So χ is a polynomial in $x_1, \dots, x_d$ alone, and it is enough to show that it does not factor as such a polynomial. (Use $x_1, \dots, x_d$ and z as coordinates. Then χ has degree zero in z, so every factor of χ does too.) It has degree exactly one in each x_i. Suppose $\chi = \chi_1 \chi_2$ with both factors nonconstant. Then each x_i appears in exactly one factor, so the variables split into two nonempty sets S and S^c, with χ_1 using only S and χ_2 only S^c. Scale the factors so that each has top coefficient 1. Comparing coefficients gives

$$m(U \cup V) = m(U)\, m(V) \qquad \text{for } U \subseteq S \text{ and } V \subseteq S^c.$$

Since G is irreducible, some cycle of positive rates meets both S and S^c. Take one with the fewest states, let W be its set of states, and put $U = W \cap S$, $V = W \cap S^c$. Expand $m(W)$ over permutations of W. The permutations that keep U and V apart sum to $m(U)m(V)$. Any other permutation with nonzero weight contains a cycle that meets both sides, which by our choice uses

all of W. A single cycle on W has sign $(-1)^{|W|-1}$ and weight $(-1)^{|W|}$ times a product of rates, so it contributes minus that product. These contributions all have the same sign, so they cannot cancel, and ours is not zero. So $m(W) \neq m(U)m(V)$, a contradiction. □

Proof of Theorem 3.1. We write Φ_1 as a sum of d exponentials on a ball where the eigenvalues of $G - \Lambda$ are far apart, do the same for Φ_1', and match the two sums term by term.

Both Φ_1 and Φ_1' are entire and agree on $\mathbb{R}^d$, so they agree on $\mathbb{C}^d$. Let $m = 2\max_i \max(q_i, q_i')$ and let B be the ball of radius $3m$ around the real point $(0, 12m, 24m, \ldots)$. On B we have $|\lambda_i - \lambda_j| > 6m$ for $i \neq j$. By Gershgorin's theorem, the matrix $G - \Lambda$ then has exactly one eigenvalue $z_i(\lambda)$ in the disc $|z + \lambda_i + q_i| \leq q_i$ for each i, and these discs are disjoint. So the z_i are simple, holomorphic and algebraic on B, and

$$\Phi_1 = \sum_i \rho_i e^{z_i}, \qquad \rho_i = \frac{n(z_i, \lambda)}{\partial_z \chi(z_i, \lambda)}, \qquad n(z, \lambda) = \mu^\top \operatorname{adj}(z + \Lambda - G)\mathbf{1},$$

where ρ_i is the residue of $n/\chi = \mu^\top (z + \Lambda - G)^{-1}\mathbf{1}$ at z_i. Since the residues of a resolvent sum to the identity, $\sum_i \rho_i = 1$. The same holds for (μ', G') with z_i', ρ_i' and χ'.

We now compare the two expansions of Φ_1 in three steps.

No mode is missing. We claim that no ρ_i vanishes identically on B. If one did, then $n(z_i(\lambda), \lambda) = 0$ on B, so n and χ would share the root z_i and hence a common factor. Since χ is irreducible (Lemma 3.3), it would divide n. But n has degree $d - 1$ in z, less than χ, and it is not zero, because its leading coefficient in z is $\sum_i \mu_i = 1$.

The modes match. On B we have $|z_i + \lambda_i| \leq m$ and $|z_j' + \lambda_j| \leq m$, so each of the differences $z_i - z_j$, $z_i' - z_j'$ and $z_i - z_j'$ stays within $2m$ of $\lambda_j - \lambda_i$. For $i \neq j$ the number $\lambda_j - \lambda_i$ moves by more than $4m$ across B. So none of these differences with $i \neq j$ is constant, and the only exponents that can differ by a constant are z_i and z_i' with the same i. Now $\sum_i \rho_i e^{z_i} - \sum_i \rho_i' e^{z_i'} = 0$ on B. Merge z_i and z_i' into one term whenever they differ by a constant, and apply Lemma 3.2. Every coefficient is then zero. If z_i were not merged, its coefficient would be ρ_i, which is not zero. So $z_i' = z_i - \kappa_i$ for a constant κ_i, the merged coefficient is $\rho_i - e^{-\kappa_i}\rho_i'$, and hence $\rho_i' = e^{\kappa_i}\rho_i$.

The constants vanish. The polynomials $\chi'(z, \lambda)$ and $\chi(z + \kappa_1, \lambda)$ share the root $z_1 - \kappa_1$ on B. The second is irreducible, so it divides the first. Both have degree d in z with leading coefficient 1, so they are equal. Their roots are the $z_i - \kappa_i$ and the $z_i - \kappa_1$. So $z_i - \kappa_i = z_j - \kappa_1$ for some j, and $j = i$ because $z_i - z_j$ is not constant for $j \neq i$. Thus $\kappa_i = \kappa_1$ for every i. Then $1 = \sum_i \rho_i' = e^{\kappa_1} \sum_i \rho_i = e^{\kappa_1}$. Also κ_1 is real. (At the center of B the matrices are real, and each disc contains one eigenvalue and is symmetric about the real axis, so z_1 and z_1' are real there.) Hence $\kappa_1 = 0$, and every $\kappa_i = 0$.

So $\chi' = \chi$ and $\rho_i' = \rho_i$, which gives $\mu'^\top (z + \Lambda - G')^{-1}\mathbf{1} = \sum_i \rho_i/(z - z_i) = \mu^\top (z + \Lambda - G)^{-1}\mathbf{1}$ on B, and so everywhere, since both sides are rational. The left and right sides are the Laplace transforms in t of Φ_t' and Φ_t. Hence $\Phi_t' = \Phi_t$ for every t, and $z = 0$ gives $r' = r$. □

4 What the counts know about a pair

This section finds what the counts know about two states at a time, namely the two flows between them up to order, and shows that this is enough to recover almost every chain.

Theorem 4.1. *Let G be irreducible and $\mu_i > 0$ for every i. The law of $L(1)$ determines μ_i and q_i for every state, and a_{ij} and c_{ij} for every pair. So by (1) it determines $\{f_{ij}, f_{ji}\}$ as an unordered pair, namely the two roots of $x^2 - c_{ij}x + \mu_i\mu_j a_{ij}$, and with it $|J_{ij}|$.*

Proof. By Theorem 3.1 the law determines r. For a set F of states let $\lambda_k \to \infty$ for every $k \notin F$. Then $e^{-\lambda \cdot L(t)}$ decreases to $e^{-\lambda \cdot L(t)}\mathbf{1}\{Y \text{ stays in } F \text{ up to } t\}$, so r tends to $r_F = \mu_F^\top(\Lambda_F - G_{FF})^{-1}\mathbf{1}$, the same function for the chain killed when it leaves F. With $x_i = \lambda_i + q_i$,

$$r_{\{i\}} = \frac{\mu_i}{x_i}, \qquad r_{\{i,j\}} = \frac{\mu_i x_j + \mu_j x_i + c_{ij}}{x_i x_j - a_{ij}}.$$

The first gives μ_i and q_i. In the second, cross-multiplying two such fractions and comparing the coefficients of $x_i x_j$ and then of x_i shows that c_{ij} and a_{ij} are determined. □

The formula for $r_{\{i,j\}}$ says what is going on. A path that jumps once between i and j leaves the same counts whether it went from i to j at time s or from j to i at time $1-s$. So one-jump paths show only the sum c_{ij}. Round trips show the product a_{ij}. A sum and a product give two numbers, but not their order.

Let S be a set of pairs. Write G^S for the matrix obtained from G by swapping f_{ij} and f_{ji} on every pair in S (so $g^S_{ij} = \mu_j g_{ji}/\mu_i$ there), with the diagonal chosen so that rows sum to zero.

Corollary 4.2 (the exit rates decide)**.** *Under the hypotheses of Theorem 4.1, every (μ', G') with the same law of $L(1)$ has $\mu' = \mu$ and $G' = G^S$ for a set S of pairs on which the net flow obeys Kirchhoff's law,*

$$\sum_{j:\{i,j\}\in S} J_{ij} = 0 \qquad \text{for every state } i. \tag{3}$$

Proof. By Theorem 4.1, G' has the same μ, the same exit rates and the same unordered pairs of flows. Let S be the set of pairs where the order differs. The exit rate of i is $\sum_j f_{ij}/\mu_i$, and swapping changes it by $-\sum_{j:\{i,j\}\in S} J_{ij}/\mu_i$. □

Corollary 4.3. *Let G be irreducible and $\mu_i > 0$ for every i.*

(a) If $d = 2$, the law of $L(1)$ determines (μ, G).

(b) If $d = 3$, it determines (μ, G) unless μ is stationary and the chain is not reversible. In that case exactly two pairs have this law, namely (μ, G) and its reversal.

(c) For every μ and almost every G, it determines (μ, G).

(d) If $d \geq 3$ and μ is the stationary law, then for almost every G exactly two pairs have this law, namely (μ, G) and its reversal.

Proof. By Corollary 4.2 we only have to find the sets S that satisfy (3). Swapping a pair with $J_{ij} = 0$ changes nothing, so we may take $J_{ij} \neq 0$ on S.

(a) A single pair with $J_{12} \neq 0$ cannot satisfy (3). So S is empty and $G' = G$.

(b) A state that lies in exactly one pair of S violates (3). With three states this leaves only S empty or S all three pairs. In the second case (3) gives $J_{12} = J_{23} = J_{31} \neq 0$. Then $\sum_j J_{ij} = 0$ at every state, so μ is stationary and the chain is not reversible. Also G^S is the reversal, and Proposition 2.1 shows that it does have the same law.

(c) Fix μ and use the flows f_{ij} as coordinates for G. For each nonempty S, condition (3) at a state that meets S is a linear equation in the flows that is not trivial. So it fails off a hyperplane, and off these finitely many hyperplanes only S empty is left.

(d) Use the coordinates (π, c, J), where J runs over the flows that obey Kirchhoff's law on the whole graph. Let S be neither empty nor everything. Some state i has a pair $\{i,j\}$ in S and a pair $\{i,k\}$ outside it. The unit flow around the triangle $i \to j \to k \to i$ is an allowed J, and it

violates (3) at i, because only its edge $\{i, j\}$ counts there. So (3) at i is again a linear equation that is not trivial, and it fails off a proper subspace. What is left is S empty or everything, which gives G and its reversal. These two differ unless $J = 0$. (For $d = 2$ every chain is reversible, so there the two coincide.) □

Remark 4.4. The proof is also a recipe. Read μ_i and q_i from the paths that never move, and c_{ij} and a_{ij} from the paths that visit two states (in both cases through the functions r_F, that is, at all horizons). Then $f_{ij} = \frac{1}{2}(c_{ij} \pm |J_{ij}|)$, and the signs are the ones for which $\sum_j f_{ij} = \mu_i q_i$ at every state. So paths that visit at most two states are enough for almost every chain.

Remark 4.5. Part (d) is sharper than "the counts cannot see irreversibility". They cannot see its direction. They do see $|J_{ij}|$ on every pair, so they know exactly how far the chain is from reversible. For long horizons it is known that the variance of L and its large deviations depend on the net flow only through even expressions [8, 12]. Here the statement is exact, at every horizon, and for the whole law.

5 Chains without a cut

This section shows that a chain with no cut has at most one rival, namely its own reversal. Theorem 4.1 reads the law through the paths that visit at most two states. Reading all of it leads to an old question of linear algebra, namely when two matrices have the same principal minors.

Proposition 5.1 (two sets of minors)**.** *Let G be irreducible. The law of $L(1)$ determines every principal minor of G and every principal minor of $G - \mathbf{1}\mu^\top$. Conversely, these minors determine the law of $L(t)$ for every t.*

Proof. The law determines the rational function r. We show that its denominator and its numerator can be read off separately, and that their coefficients are the two sets of minors.

By Theorem 3.1 the law determines $r = n_0/\chi_0$, where

$$\chi_0(\lambda) = \det(\Lambda - G), \qquad n_0(\lambda) = \mu^\top \operatorname{adj}(\Lambda - G)\mathbf{1}.$$

The polynomial χ_0 is irreducible (the proof of Lemma 3.3 with $z = 0$). The polynomial n_0 is not zero and has smaller degree, so χ_0 does not divide it. Hence the fraction is in lowest terms, and r determines χ_0 and n_0 up to a common constant factor. The factor is fixed because the coefficient of $\lambda_1 \cdots \lambda_d$ in χ_0 is 1. (A second pair (μ', G') with the same law has $r' = r$, with a denominator of the same degree and the same top coefficient. So $\chi_0' = \chi_0$ and $n_0' = n_0$, and we do not need G' to be irreducible.)

Now we read the coefficients. For a matrix A we have $\det(\Lambda - A) = \sum_U \det(-A_{UU}) \prod_{i \notin U} \lambda_i$. With $A = G$ this is χ_0, so the coefficients of χ_0 are the principal minors of $-G$, which are those of G up to the sign $(-1)^{|U|}$. Next, the rule for the determinant of a rank one change gives $\det(\Lambda - G + \mathbf{1}\mu^\top) = \chi_0 + n_0$. So the coefficients of $\chi_0 + n_0$ are the principal minors of $-(G - \mathbf{1}\mu^\top)$.

For the converse, the two sets of minors give χ_0 and $\chi_0 + n_0$, and so r. Since the coordinates of $L(t)$ sum to t, we have $\Phi_t(\lambda + z\mathbf{1}) = e^{-zt}\Phi_t(\lambda)$. So $z \mapsto r(\lambda + z\mathbf{1})$ is the Laplace transform of $t \mapsto \Phi_t(\lambda)$, and it determines $\Phi_t(\lambda)$ for every t. □

For example, the sets U with one element give q_i from the first set of minors and $q_i + \mu_i$ from the second. So the second set is where the start law enters.

Two things always keep the principal minors of a matrix A, namely passing to DAD^{-1} for a diagonal matrix D and passing to the transpose. For a rate matrix these are familiar. The

transpose with the right D is the reversal, since $\hat{G} = \Pi^{-1} G^{\top} \Pi$ with $\Pi = \operatorname{diag}(\pi)$. Say that G has a *cut* if there is a set X of states with $2 \le |X| \le d-2$ such that both blocks G_{XX^c} and G_{X^cX} have rank at most one. (So a chain on two or three states has no cut.) Loewy [17] proved that if A is irreducible, has at least four rows and has no cut, then every matrix with the same principal minors as A is DAD^{-1} or $DA^{\top}D^{-1}$ for some diagonal D. For two and three rows the same holds for every irreducible A, by Hartfiel and Loewy [11] (see [5, Lemma 2.8]). We refer to these two statements together as the theorem quoted above.

Theorem 5.2 (at most two candidates)**.** *Let G be irreducible with no cut, and let π be its stationary law. Suppose (μ', G') gives the same law of $L(1)$ as (μ, G). Then $\mu' = \mu$, and G' is either G or its reversal $\hat{G}$. Moreover $(\mu, \hat{G})$ does give the same law if and only if, for the stationary chain with rates G and $w_i = \mu_i/\pi_i$,*

$$\mathbb{E}_{\pi}\big[w(Y_0) \mid L(1)\big] = \mathbb{E}_{\pi}\big[w(Y_1) \mid L(1)\big]. \tag{4}$$

In words, a chain with no cut has at most one rival, its own reversal, and the rival passes exactly when the counts cannot tell the first state from the last as far as the weight w is concerned. A stationary start has $w = 1$, and then (4) is clear.

Proof. The first part is linear algebra. The theorem quoted above leaves two families of matrices, and only one member of each family is a rate matrix. The second part runs the film backward.

The candidates. By Proposition 5.1, G' has the same principal minors as G. By the theorem quoted above, $G' = DGD^{-1}$ or $G' = DG^{\top}D^{-1}$ for an invertible diagonal matrix D. We use that the rows of G' sum to zero, that is, $G'\mathbf{1} = 0$. In the first case this says $GD^{-1}\mathbf{1} = 0$. Since G is irreducible, the only vectors it sends to zero are the multiples of $\mathbf{1}$. So D is a multiple of the identity and $G' = G$. In the second case $G^{\top}D^{-1}\mathbf{1} = 0$. The only vectors that $G^{\top}$ sends to zero are the multiples of π, so D^{-1} is a multiple of Π and $G' = \Pi^{-1}G^{\top}\Pi = \hat{G}$.

The start. By the proof of Proposition 5.1 the law determines the polynomial n_0, and the coefficient of $\prod_{k \ne i} \lambda_k$ in n_0 is μ_i. So $\mu' = \mu$.

When the reversal passes. For a bounded function F we compare $\mathbb{E}F(L(1))$ under (μ, G) and under $(\mu, \hat{G})$. Write $\mathbb{E}_{\pi}$ for the stationary chain with rates G. The chain (μ, G) is this chain with its start reweighted, so

$$\mathbb{E}_{(\mu,G)}F(L(1)) = \mathbb{E}_{\pi}\big[w(Y_0)F(L(1))\big].$$

In the same way the chain $(\mu, \hat{G})$ is the stationary chain with rates $\hat{G}$ weighted by w at its first state (both chains have the stationary law π). Run the film backward, as in Proposition 2.1. The stationary chain with rates $\hat{G}$ becomes the stationary chain with rates G, its first state becomes the last state Y_1, and $L(1)$ does not change. So

$$\mathbb{E}_{(\mu,\hat{G})}F(L(1)) = \mathbb{E}_{\pi}\big[w(Y_1)F(L(1))\big].$$

The two right sides agree for every F exactly when $w(Y_0) - w(Y_1)$ has conditional mean zero given $L(1)$, which is (4). □

A cut asks for 2×2 minors to vanish, so almost every rate matrix has none. Thus Theorem 5.2 names the exceptional chains in parts (c) and (d) of Corollary 4.3. In part (c) an exceptional chain has a cut or satisfies (4). In part (d) it has a cut or is reversible (then the two candidates are the same chain). In part (c) the second case can happen with a start that is not stationary, as we will see after Theorem 7.1. For a chain with every rate drawn at random it does not happen (`check_rigidity.py`).

The same two operations, and the same theorem of Loewy, appear for the kernels of permanental processes, which describe local times of Markov processes at certain random times [19]. What is different here is the fixed horizon and the start law, which bring in the second matrix $G - \mathbf{1}\mu^\top$.

6 The counts measure the entropy production

A stationary start hides the arrow of time. One might guess that the counts then say nothing about how irreversible the chain is. The opposite is true.

The *entropy production rate* of the chain at time 0 is

$$\sigma(\mu, G) = \sum_{i<j} (f_{ij} - f_{ji}) \log \frac{f_{ij}}{f_{ji}}, \qquad f_{ij} = \mu_i g_{ij}$$

(see [24]). A pair with $f_{ij} = f_{ji} = 0$ contributes nothing. Each term is nonnegative (and may be infinite if only one of the two flows vanishes), and $\sigma = 0$ exactly when $J = 0$. For a stationary start, $t\sigma$ is the relative entropy of the law of the path on $[0, t]$ with respect to the law of the same path run backward. So σ measures how far the film is from its reverse. For three states i, j, k with all six rates positive, the *affinity* of the triangle is

$$A_{ijk} = \log \frac{g_{ij} g_{jk} g_{ki}}{g_{ik} g_{kj} g_{ji}},$$

the logarithm of the odds of going around the triangle one way and not the other.

Theorem 6.1 (irreversibility)**.** *Let G be irreducible, and suppose that $g_{ij} > 0$ exactly when $g_{ji} > 0$.*

(a) If $\mu_i > 0$ for every i, the law of $L(1)$ determines $\sigma(\mu, G)$. With the pair data of Theorem 4.1,

$$\sigma(\mu, G) = \sum_{i<j} |J_{ij}| \log \frac{c_{ij} + |J_{ij}|}{c_{ij} - |J_{ij}|}, \qquad |J_{ij}| = \sqrt{c_{ij}^2 - 4\mu_i \mu_j a_{ij}},$$

where the sum is over the pairs with $a_{ij} > 0$.

(b) For every start law, the law of $L(1)$ determines $|A_{ijk}|$ for every triangle with all six rates positive. So if every rate is positive, the law of $L(1)$ decides whether the chain is reversible.

In words, the counts know the entropy production exactly, whether or not they can tell which way the film runs. With a stationary start they cannot tell, and the entropy production then says how different the two directions are. Each pair of states contributes its own share. Each triangle with all six rates positive shows the size of the force that drives the chain around it, but not its sign.

Proof. Both parts use the same remark. The counts know a sum and a product, and the quantity we want does not change when the two numbers are swapped.

(a) Each term of σ does not change when f_{ij} and f_{ji} are swapped. So σ is a function of the unordered pairs $\{f_{ij}, f_{ji}\}$, which the law determines by Theorem 4.1. For the formula, let $a_{ij} > 0$. The two flows are $\frac{1}{2}(c_{ij} \pm |J_{ij}|)$, both positive, so their difference is $\pm|J_{ij}|$ and the logarithm of their ratio has the same sign. The value of $|J_{ij}|$ is the one in (1). A pair with $a_{ij} = 0$ has both flows zero by our assumption on G, and contributes nothing.

(b) Here the two numbers are the weights of the two ways around a triangle, namely $\omega = g_{ij} g_{jk} g_{ki}$ and $\bar{\omega} = g_{ik} g_{kj} g_{ji}$, so that $A_{ijk} = \log(\omega/\bar{\omega})$. By Proposition 5.1 the law determines the principal

minors of $-G$, for every start law. Those of order one are the q_i, and those of order two give $a_{ij} = q_i q_j - \det(-G_{\{i,j\}\{i,j\}})$. For order three we expand the determinant over the six permutations. The identity gives $q_i q_j q_k$, each transposition gives one term $-q\,a$, and the two cyclic permutations give $-\omega$ and $-\bar{\omega}$. So

$$\det(-G_{UU}) = q_i q_j q_k - q_i a_{jk} - q_j a_{ik} - q_k a_{ij} - (\omega + \bar{\omega}), \qquad U = \{i, j, k\}.$$

Hence the law determines the sum $\omega + \bar{\omega}$. It also determines the product $\omega\bar{\omega} = a_{ij}a_{jk}a_{ki}$. A sum and a product give the unordered pair $\{\omega, \bar{\omega}\}$, and with it $|A_{ijk}| = |\log(\omega/\bar{\omega})|$.

Now let every rate be positive. We show that the chain is reversible exactly when $\omega = \bar{\omega}$ for every triangle, which the law decides by what we just proved. If the chain is reversible then $\pi_i g_{ij} = \pi_j g_{ji}$ on every pair, and multiplying three of these gives $\omega = \bar{\omega}$. Conversely, suppose $\omega = \bar{\omega}$ for every triangle. Put $\theta_1 = 1$ and $\theta_i = g_{1i}/g_{i1}$, so that $\theta_1 g_{1i} = \theta_i g_{i1}$. For $i, j \neq 1$ the triangle $\{1, i, j\}$ gives $g_{1i}g_{ij}g_{j1} = g_{1j}g_{ji}g_{i1}$, that is, $\theta_i g_{ij} = \theta_j g_{ji}$. So θ is a reversible measure. (This is Kolmogorov's criterion, see [13].) □

Part (a) does not ask for a chain without a cut. By Corollary 4.2 two chains with the same counts have the same unordered pairs of flows, so they have the same σ, whatever the reason for the tie. This is also why data that cannot see the direction of time can give σ exactly. The data do not single out the chain, but σ takes the same value at every chain they allow.

Remark 6.2. (1) Maes, Netočný and Wynants [18] showed that, close to a reversible chain and to leading order, the rate function for the large deviations of the occupation times is a quarter of the excess of the entropy production over its stationary value. Part (a) is exact, at a single finite horizon, and needs no closeness.

(2) Write $s = J_{ij}^2/c_{ij}^2$. The term of the pair in (a) is $c_{ij}\,\varphi(s)$ with $\varphi(s) = \sqrt{s}\log\frac{1+\sqrt{s}}{1-\sqrt{s}} = 2s + \frac{2}{3}s^2 + \cdots$, an analytic function of s for $|s| < 1$. So σ is a smooth function of c and J^2, the quantities the counts see. Near a reversible chain the counts lose the net flow J itself (we come back to this in Theorem 8.2), but nothing of this kind happens to σ.

(3) Part (a) uses only the functions r_F for faces F of at most two states, that is, the paths that stay on at most two states (at all horizons, which one horizon gives by Theorem 3.1). Part (b) uses only the faces of at most three states, as long as the start gives positive mass to some state of the triangle. (The minors of G inside a set F are the coefficients of the denominator of r_F from the proof of Theorem 4.1, and r_F is zero when no path starts in F.)

7 Lumping gives more ties

A chain with a cut can have other rivals, and a start that is not stationary does not save us. The reason is that a chain can contain a smaller chain to which Proposition 2.1 applies.

Call a set M of states a *module* if, with O its complement,

(i) every state of M leaves to $x \in O$ at the same rate, $g_{bx} = \gamma_x$,

(ii) every entrance into M lands the same way, $g_{xb} = r_x \nu_b$ for a law ν on M,

(iii) the start inside M follows ν, namely $\mu_b = \mu(M)\nu_b$.

From outside, M then looks like one state $*$. The *lumped chain* lives on $O \cup \{*\}$ with rates g_{xy} inside O, rates r_x into $*$ and γ_x out of it, and start law $\bar{\mu}$ (with $\bar{\mu}_* = \mu(M)$). The *inside chain* lives on M with the rates $g_{bb'}$. By (i) and (ii) the blocks G_{MO} and G_{OM} have rank one, so a module with two or more states on each side is a cut.

Theorem 7.1 (two reversals)**.** *Let M be a module.*

(a) (Reverse the outside, keep the inside.) Suppose $\bar{\mu}$ is stationary for the lumped chain and positive at every state. Reverse the lumped chain with respect to $\bar{\mu}$, and let G' be the chain with the reversed rates on O, the reversed rates $\hat{r}_x \nu_b$ into M and $\hat{\gamma}_x$ out of M, and the old rates $g_{bb'}$ inside M. Then (μ, G') and (μ, G) give the same law of $L(t)$ for every t.

(b) (Reverse the inside, keep the outside.) Suppose ν is stationary for the inside chain and positive at every state of M. Let G'' be G with the inside chain replaced by its reversal with respect to ν, namely $g''_{bb'} = \nu_{b'} g_{b'b} / \nu_b$ for $b, b' \in M$. Then (μ, G'') and (μ, G) give the same law of $L(t)$ for every t.

The idea is that a chain with a module is two films, one inside the other. The outer film is the lumped chain. Each of its visits to $*$ holds an inner film, a piece of the inside chain. The counts add up times and forget the order of things, so we may run the outer film backward and leave the inner films alone, or run each inner film backward and leave the outer one alone.

Proof. We build Y in two steps. First run the lumped chain Z from $\bar{\mu}$ up to time t. Then, for each visit of Z to $*$, fill the visit with a fresh copy of the inside chain (rates $g_{bb'}$ on M, started from ν), run for the length of the visit. The fillings are independent of each other and of Z.

We claim that this builds the chain (μ, G). By (i) a path of Y inside M leaves to x at the rate γ_x wherever it is. So the time and place of the exit do not depend on what happens inside M, and until the exit the path moves by the inside rates. By (ii) every entrance into M lands by ν, and by (iii) so does a start inside M. Hence every visit starts from ν.

The time that Y spends in a state of O is the time that Z spends there. The time that Y spends in a state b of M is the sum over the visits of the time the filling spends in b. So the law of $L(t)$ depends on two things only. The first is the joint law of the times Z spends in the states of O and the list of lengths of its visits to $*$, where the order of the list does not matter. The second is the law of the times a filling of a given length spends in the states of M.

(a) Run the film of Z backward. Since $\bar{\mu}$ is stationary, this is the reversed lumped chain started from $\bar{\mu}$. It has the same times in O and the same visit lengths, in the opposite order. So the first of the two things is the same for Z and for its reversal, and we fill the visits as before. The result is built from the reversed lumped chain and the old inside chain in the same two steps, so it is the chain (μ, G').

(b) Keep Z and run the film of each filling backward. A filling is the inside chain started from its stationary law ν and run for a fixed time. So its film run backward is the reversed inside chain started from ν, and it spends the same time in each state of M. So the second of the two things is the same for the inside chain and for its reversal. The result is built from Z and the reversed inside chain in the same two steps, so it is the chain (μ, G''). □

A single state is always a module, and then (a) is Proposition 2.1. Part (b) asks nothing of the start outside M or of the lumped chain. On the level of the polynomial χ_0 these ties are known. Every matrix with the same principal minors as an irreducible matrix is reached from it by diagonal similarity, the transpose, and transposing one side of a cut [11, 3, 5], and (a) and (b) are of this kind. What Theorem 7.1 adds is that they pass the second test of Proposition 5.1, the one that involves the start.

The second test matters. Take a rate matrix with a cut X such that neither X nor its complement is a module (the blocks G_{XX^c} and G_{X^cX} have rank one, but the rows of G_{XX^c} are not equal). Transposing one side of the cut and rescaling gives a rate matrix G' with the same

principal minors as G, and so with the same stationary law (it is proportional to the principal minors of order $d-1$ of $-G$). In the three random examples of this kind that we tried, the minors of $G' - \mathbf{1}\pi^\top$ differ from those of $G - \mathbf{1}\pi^\top$, and the counts differ, by $2 \cdot 10^{-4}$ to $9 \cdot 10^{-4}$ in the Laplace transform (`check_cut_transpose.py`). This is an observation. We have not proved that the counts differ for almost every such chain, and for special rates they need not. For the chains of Example 7.3 the same script confirms that G' is exactly the transpose of G across the module, up to a diagonal similarity.

Example 7.2 (the reversal can win without a stationary start)**.** Let M be a module with three states and let O be a single state z, with every rate positive, and let ν be stationary for the inside chain. Here M is not a cut, since $|O| = 1$, and for a typical choice of the rates G has no cut at all. We claim that $G'' = \hat{G}$. First, the stationary law π of G is proportional to ν on M. To see this, let π be ν on M times a constant, with π_z chosen so that $\pi_z r_z = \pi(M)\gamma_z$ (the flow out of z equals the flow into z). At a state b of M the flow through the inside balances because ν is stationary there, and the flow from z, namely $\pi_z r_z \nu_b$, equals the flow to z, namely $\pi_b \gamma_z$. So this π is stationary. Second, we just saw that $\pi_z g_{zb} = \pi_b g_{bz}$ for every b. So reversal with respect to π keeps the rates between z and M, and inside M it is reversal with respect to ν. This is G''.

The start $\mu = a\nu + (1-a)\delta_z$ makes M a module for every $0 < a < 1$. So by (b) it gives the same counts under G and under $\hat{G}$, and it is stationary for only one value of a. Thus (4) can hold with w not constant. Here w takes one value on M and another at z.

Example 7.3 (four states)**.** Take $O = \{1, 2\}$ and $M = \{3, 4\}$. Let the lumped chain on $\{1, 2, *\}$ have rates $g_{12} = 0.9$, $g_{21} = 0.3$, $r_1 = 0.5$, $r_2 = 1.1$, $\gamma_1 = 0.8$, $\gamma_2 = 0.4$, and let $\bar{\mu}$ be its stationary law. Take $\nu = (0.35, 0.65)$ and inside rates $g_{34} = 1.7$, $g_{43} = 0.2$. Every rate of G is positive, and μ is not stationary for G, because ν is not stationary for the inside. The lumped chain is not reversible, so $G' \neq G$ (their entries differ by up to 0.59). The two laws of $L(t)$ agree to 10^{-15} at $t = 0.3$, 1 and 4. In the language of Corollary 4.2, S is the five pairs that are not inside M. By Theorem 6.1 the two chains also have the same entropy production rate, $\sigma = 0.7325$ (Figure 1).

When both parts apply, and neither the inside chain nor the lumped chain is reversible, we get four chains with the same counts, namely G, the chain with only the outside reversed, the chain with only the inside reversed, and the chain with both reversed (checked on six states with a stationary start, where the last one is $\hat{G}$).

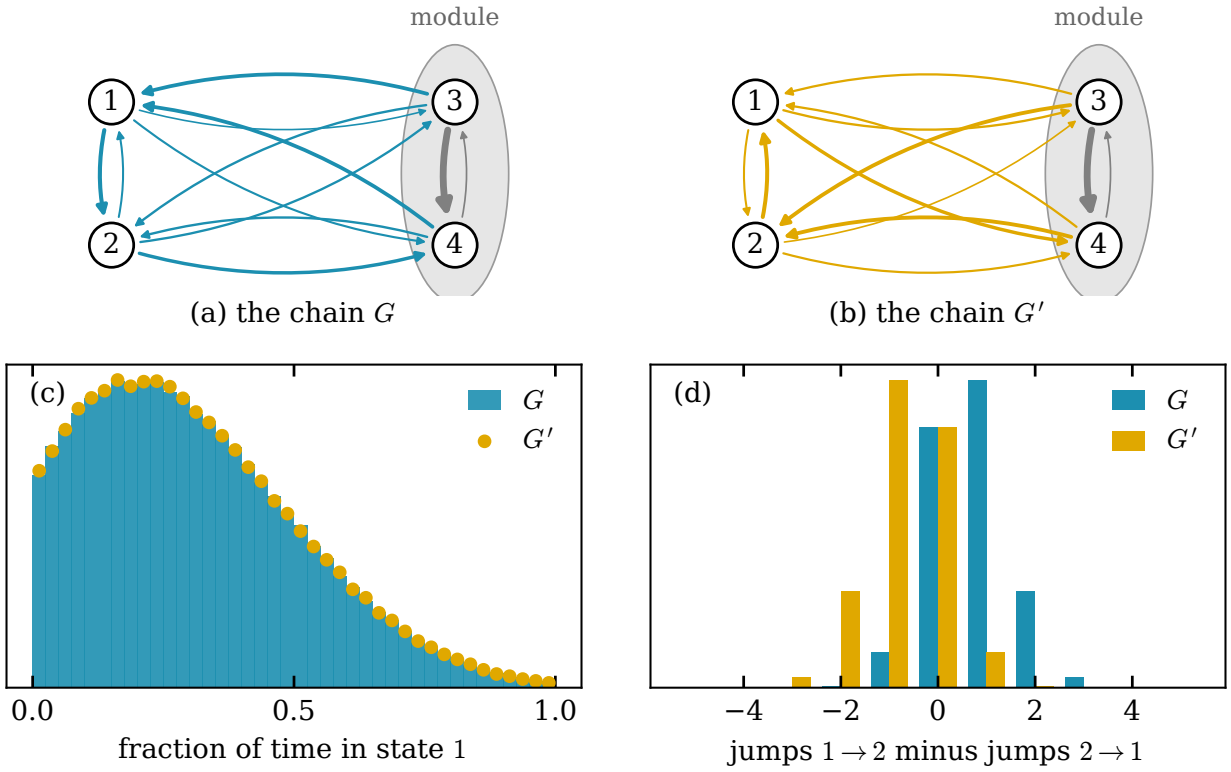
8 What is lost

This section compares the information in the counts with the information in the full path, first when switches are rare, then at a reversible chain, and then over a long horizon.

Now let $G = TQ$, with μ known and positive and every $Q_{ij} > 0$. We compare the Fisher information about Q in the full path on $[0, 1]$, written I_{path}, with the information in $L(1)$, written I_L. Here $c_{ij} = \mu_i Q_{ij} + \mu_j Q_{ji}$ and $q_i = \sum_j Q_{ij}$ are functions of Q, and for three states we put

$$\tau_{ijk} = \sum_{(x,y,z)} \mu_x Q_{xy} Q_{yz} = \sum_y \frac{c_{xy} c_{yz} + J_{xy} J_{yz}}{2\mu_y},$$

where the first sum is over the six orders of $\{i, j, k\}$ and in the second x and z are the two states other than y.



(a) the chain G (b) the chain G'

Figure 1: The two chains of Example 7.3. (a), (b) The rates, with thicker arrows for larger rates. The gray arrows inside the module are the same in both chains, and every other rate differs. (c) The fraction of time in state 1 up to time $t = 4$, from $4 \cdot 10^5$ simulated paths of each chain. The two histograms agree, as they must (the atoms at 0 and 1 are left out). (d) The number of jumps from 1 to 2 minus the number from 2 to 1 on the same paths. This sees the order of the visits, and it tells the chains apart at once.

Theorem 8.1 (rare switches)**.** *As $T \to 0$,*

$$I_{\text{path}} = T \operatorname{diag}\Big(\frac{\mu_i}{Q_{ij}}\Big) + O(T^2), \qquad I_L = T \sum_{i<j} \frac{\nabla c_{ij} \nabla c_{ij}^\top}{c_{ij}} + O(T^2).$$

So the efficiencies (the eigenvalues of $I_{\text{path}}^{-1} I_L$) split in half, with $\binom{d}{2}$ of them tending to 1 and $\binom{d}{2}$ tending to 0. The counts are first-order efficient for every two-way flow c_{ij} and carry no first-order information about any net flow. For a direction v with $v \cdot \nabla c_{ij} = 0$ for all pairs,

$$v^\top I_L v = T^2 \Big[\sum_i \mu_i (v \cdot \nabla q_i)^2 + \sum_{i<j<k} \frac{(v \cdot \nabla \tau_{ijk})^2}{2\tau_{ijk}} \Big] + O(T^3). \tag{5}$$

Proof. The score of the path for Q_{ij} is $s_{ij} = N_{ij}/Q_{ij} - TL_i$, where N_{ij} counts the jumps from i to j, and $I_L = \mathbb{E}[\bar{s}\bar{s}^\top]$ with $\bar{s} = \mathbb{E}[s \mid L]$. Since $\mathbb{E}N_{ij} = T\mu_i Q_{ij} + O(T^2)$, the first claim follows. For the second we sort by the face of the simplex that holds L. The chance of three or more jumps is $O(T^3)$, and so is the second moment of the score on that event.

A vertex. The event $L = e_i$ means no jump, so $\bar{s} = -T\nabla q_i$ there, with probability $\mu_i + O(T)$. This contributes $T^2 \mu_i \nabla q_i \nabla q_i^\top$.

An edge. Given L on the edge of i and j, up to a relative error $O(T)$ that is uniform along the edge, the path made one jump, from i to j with probability f_{ij}/c_{ij} and from j to i otherwise. (A round trip also lands on the edge, with a density of order T^2 against order T for one jump.) So $\bar{s}_{ij} = \mu_i/c_{ij} + O(T)$ and $\bar{s}_{ji} = \mu_j/c_{ij} + O(T)$, that is, $\bar{s} = \nabla c_{ij}/c_{ij} + O(T)$. The edge has probability $Tc_{ij} + O(T^2)$. This contributes $T\nabla c_{ij}\nabla c_{ij}^\top/c_{ij} + O(T^2)$, and $O(T^3)$ in a direction v with $v \cdot \nabla c_{ij} = 0$.

A triangle. Given L in the open face of i, j and k, up to a uniform relative error $O(T)$ the path made two jumps in one of six orders. Each order fits every point of the face, and the order (x, y, z) has probability $\mu_x Q_{xy} Q_{yz}/\tau_{ijk}$. Since τ_{ijk} is linear in each rate, $\bar{s} = \nabla\tau_{ijk}/\tau_{ijk} + O(T)$. The face has probability $T^2\tau_{ijk}/2 + O(T^3)$, because the two jump times fill half of the unit square. This contributes the last term of (5).

For the efficiencies, in the coordinates f_{ij} the matrix $I_{\text{path}}^{-1} I_L$ tends to a block for each pair, equal to $\operatorname{diag}(f_{ij}, f_{ji})$ times the all-ones matrix over c_{ij}, with eigenvalues 1 and 0. □

When switches are rare, a typical path that tells us anything has one jump. The counts then say which two states and when, but not which way. This is why they keep the two-way flows and lose the net flows. The net flows come back at the next order from two places in (5), the paths that never move (their number gives the exit rates) and the paths that visit three states.

Theorem 8.2 (stationary start)**.** *Let the start be the stationary law, and use the coordinates (π, c, J) for Q, where J obeys Kirchhoff's law at every state. Then the law of $L(t)$ is an even function of J. So at every reversible chain, and for every t, the information matrix of $L(t)$ vanishes on the J directions, a space of dimension $\binom{d-1}{2}$.*

Proof. Reversal sends (π, c, J) to $(\pi, c, -J)$ and keeps the law (Proposition 2.1). The derivative of an even function vanishes at 0, so the score in every J direction is zero at $J = 0$. The J directions are the cycle space of the complete graph, of dimension $\binom{d}{2} - (d-1)$. □

Remark 8.3. Near a reversible chain the law depends on J through quadratic terms such as the one in τ_{ijk}. So from n copies of the counts one should expect to learn J^2 at the usual rate, hence $|J|$ only at rate $n^{-1/4}$ near $J = 0$, and never its sign. Theorem 9.1 below proves that no estimator of $|J|$ does better than $n^{-1/4}$. (Rates of this kind are known for models whose information matrix loses rank by one [21], which is the case of three states. There the model is assumed identifiable, and ours is not, because of the sign.) Numerically, in one example with three states and $T = 2$, the smallest eigenvalue of the information matrix is $0.94\, j^2$, $0.96\, j^2$ at $j = 0.04$, 0.02, where j is the flow around the triangle.

The crossings of Paper C sit at the other end, where T is of order $\log N$ and so is large. There the counts keep the stationary law and, as far as we can see, nothing else. The first half of this is a theorem. Let π be the stationary law of Q, let $Z = (\mathbf{1}\pi^\top - Q)^{-1} - \mathbf{1}\pi^\top$, and let $\Gamma = \operatorname{diag}(\pi)Z + Z^\top \operatorname{diag}(\pi)$, the asymptotic covariance of the occupation times (so $\operatorname{Cov} L(1) = \Gamma/T + O(T^{-2})$ for the generator TQ).

Theorem 8.4 (long horizons)**.** *Let Q be irreducible with every $Q_{ij} > 0$, let the start be any fixed law or the stationary law, and let $D = \operatorname{diag}(\pi_i/Q_{ij})$. Write π' for the matrix of the derivatives $\partial\pi_k/\partial Q_{ij}$.*

(a) The identity $\pi' D^{-1} \pi'^\top = \Gamma$ holds. In words, to leading order in $1/T$ the Cramér–Rao bound of the full path for π equals the covariance matrix of the occupation fractions.

(b) As $T \to \infty$, $I_{\mathrm{path}} = TD + O(1)$ and $I_L \succeq T\,\pi'^{\top}\Gamma^{+}\pi' - O(1)$, where Γ^{+} is the inverse of Γ on the vectors with sum zero. So $d-1$ of the efficiencies tend to 1.

Proof. (a) We compute both sides as quadratic forms in a vector η and find the same sum over pairs. We use the three facts $Z\mathbf{1} = 0$, $\pi^{\top}Z = 0$ and $-QZ = I - \mathbf{1}\pi^{\top}$, which follow from $(\mathbf{1}\pi^{\top} - Q)(Z + \mathbf{1}\pi^{\top}) = I$. Differentiating $\pi^{\top}Q = 0$ gives $d\pi^{\top}Q = -\pi^{\top}dQ$. Multiply by Z on the right and use $d\pi^{\top}\mathbf{1} = 0$. This gives $d\pi^{\top} = \pi^{\top}(dQ)Z$. Raising Q_{ij} (and lowering Q_{ii}) gives $\partial\pi/\partial Q_{ij} = \pi_i(Z_{j\cdot} - Z_{i\cdot})$. So for a vector η and $u = Z\eta$, the entry of $\pi'^{\top}\eta$ at (i,j) is $\pi_i(u_j - u_i)$, and

$$\eta^{\top}\pi' D^{-1}\pi'^{\top}\eta = \sum_{i\neq j} \frac{Q_{ij}}{\pi_i}\,\pi_i^2(u_j-u_i)^2 = \sum_{i\neq j}\pi_i Q_{ij}(u_j-u_i)^2.$$

On the other side $\eta^{\top}\Gamma\eta = 2\sum_i \pi_i\eta_i u_i$. Here $-Qu = \eta - (\pi\cdot\eta)\mathbf{1}$ and $\pi\cdot u = 0$, so $\eta^{\top}\Gamma\eta = -2\sum_i \pi_i u_i (Qu)_i$. Now expand the squares in the display. The terms with u_i^2 and with u_j^2 each give $\sum_i \pi_i q_i u_i^2$ (the second by $\pi^{\top}Q = 0$), and the cross terms give $-2\sum_i \pi_i u_i\big((Qu)_i + q_i u_i\big)$. The sum is the same number.

We note for part (b) that Γ is positive definite on the vectors with sum zero. Indeed the display vanishes only if u is constant. Then $u = 0$ because $\pi\cdot u = 0$, and so $\eta = (\pi\cdot\eta)\mathbf{1}$.

(b) The first claim is as in Theorem 8.1. The path information is diagonal with entries $T\,\mathbb{E}L_i(1)/Q_{ij}$, and $\mathbb{E}L_i(1) = \pi_i + O(1/T)$. (A stationary start adds the information in the first state, which is $O(1)$.)

For the second claim we bound I_L from below by what the mean of L alone tells us. Let m and V be the mean and the covariance of $L = L(1)$ under the generator TQ, and let s be the score of L. Differentiating $\mathbb{E}L = m$ gives $\mathbb{E}[(L-m)s^{\top}] = m'$. So the covariance matrix of the pair (L,s) has the blocks V, m' and I_L, and it is positive semidefinite. Its Schur complement gives $I_L \succeq m'^{\top}V^{+}m'$, where V^{+} is the pseudoinverse of V. (The coordinates of L sum to 1, so $V\mathbf{1} = 0$ and V^{+} lives on the vectors with sum zero.) Now $e^{tQ} - \mathbf{1}\pi^{\top}$ decays exponentially, locally uniformly in Q and together with its derivatives in Q. Since $m^{\top} = \pi^{\top} + \frac{1}{T}\mu^{\top}\int_0^T (e^{uQ} - \mathbf{1}\pi^{\top})\,du$, and this integral and its derivative in Q stay bounded as T grows,

$$m = \pi + O(1/T), \qquad m' = \pi' + O(1/T), \qquad V = \Gamma/T + O(T^{-2}).$$

Since Γ is invertible on the vectors with sum zero, so is V for large T, with $V^{+} = T\,\Gamma^{+} + O(1)$. This gives the bound.

Finally we count the efficiencies that tend to 1. The nonzero eigenvalues of $D^{-1}\pi'^{\top}\Gamma^{+}\pi'$ are those of $\Gamma^{+}\pi' D^{-1}\pi'^{\top}$, which is $\Gamma^{+}\Gamma$ by (a). So they are 1 with multiplicity $d-1$. The eigenvalues of $I_{\mathrm{path}}^{-1}I_L$ can only go up when I_L does, so by the two estimates of (b) the largest $d-1$ of them are at least $1 - O(1/T)$. Every efficiency is at most 1, because L is a function of the path. □

So over a long horizon the counts estimate the stationary law as well as the full path does. That the occupation fractions are an efficient estimate of π is classical (see [2] for the path likelihood and [10] for efficiency). Part (a) is the identity behind it. The other $(d-1)^2$ efficiencies appear to tend to 0 (Table 1 and Figure 2), but we have not proved this. For the chain of Table 1, three of them fall off like $1/T$ and the smallest like T^{-2}.

9 How fast the counts learn the net flow

Suppose we are given n independent copies of the counts of a stationary chain. Theorem 8.2 says that at a reversible chain the counts carry no first-order information about the net flow. Here we

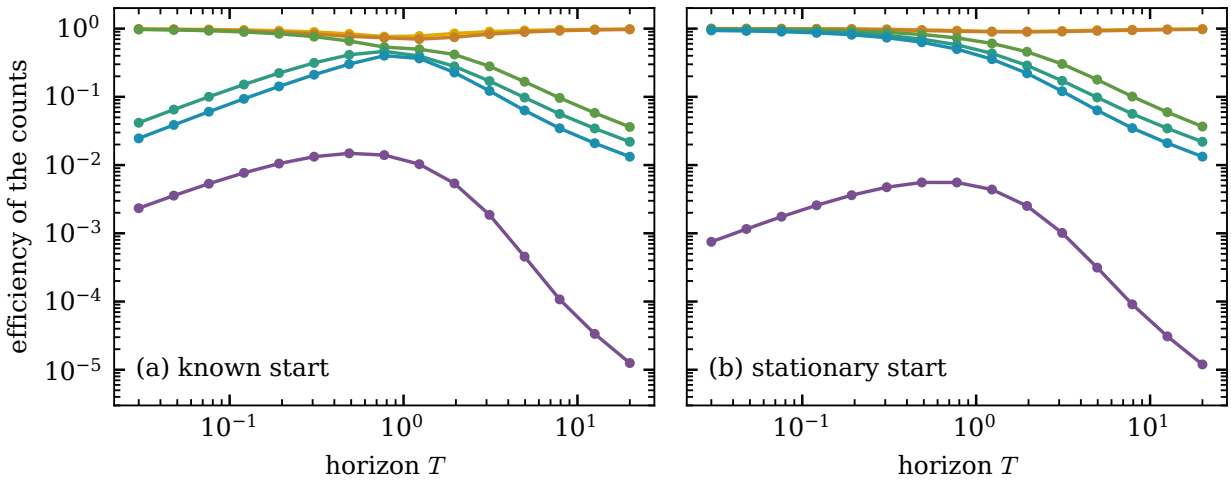


Figure 2: The six efficiencies of the counts against the horizon T, for the chain of Table 1. For few switches they split as in Theorem 8.1. Three are lost with a known start, and only the flow around the triangle with a stationary start. For many switches two tend to 1 (Theorem 8.4), three fall off like $1/T$ and the smallest like T^{-2} (between $T = 10$ and $T = 20$ their slopes are -1.04, -0.99, -1.00 and -2.2 with a known start, and -1.06, -0.99, -1.01 and -2.1 with a stationary start).

T	known start	stationary start
0.05	0.98, 0.97, 0.95, 0.07, 0.04, 0.004	1.00, 1.00, 0.98, 0.95, 0.92, 0.001
1	0.75, 0.72, 0.50, 0.43, 0.42, 0.012	0.92, 0.90, 0.67, 0.50, 0.42, 0.005
20	0.99, 0.97, 0.04, 0.02, 0.01, 10^{-5}	0.99, 0.98, 0.04, 0.02, 0.01, 10^{-5}

Table 1: The six efficiencies of the counts for one chain on three states (computed without simulation for the N-step chain with N at least 240, which approximates the chain in continuous time). For small T three are lost with a known start, as in Theorem 8.1, and only the flow around the triangle with a stationary start. For large T only two survive (the stationary law, Theorem 8.4). This last regime is the one of Paper C's crossings, where T is of order $\log N$.

make this a statement about estimators. A small net flow of size δ changes the law of the path by an amount of order δ, but it changes the law of the counts only by an amount of order δ^2. So the counts need at least of the order of δ^{-4} copies to notice it, where for the path, which is a regular model in δ, one expects δ^{-2}.

Let (π, c) be a reversible chain with every $\pi_i > 0$, with rates $g^0_{ij} = c_{ij}/(2\pi_i)$, and let $j = (j_{ij})$ be a *circulation*, namely $j_{ji} = -j_{ij}$, $\sum_k j_{ik} = 0$ for every i, and $j_{ij} = 0$ whenever $c_{ij} = 0$. Put $b_{ij} = j_{ij}/c_{ij}$ on the pairs with $c_{ij} > 0$, and $\bar{b} = \max|b_{ij}| > 0$. For $|\delta|\bar{b} < 1$ let G_δ be the chain with rates

$$g^\delta_{ij} = \frac{c_{ij} + \delta j_{ij}}{2\pi_i} = g^0_{ij}(1 + \delta b_{ij}).$$

It has the same stationary law π, the same exit rates and the same two-way flows as the reversible chain, and its net flow is $J = \delta j$. Let P_δ be the law of $L(t)$ under G_δ with start π, and let N be the number of jumps of the reversible chain up to time t.

Theorem 9.1 (the counts are slow to see a net flow)**.** *Put* $K = \frac{1}{4}\bar{b}^4\,\mathbb{E}_0\big[N^4(9/4)^N\big]$, *which is finite.*

For $|\delta|\bar{b} \le \frac{1}{2}$,

$$\chi^2(P_\delta, P_0) = \int \Big(\frac{dP_\delta}{dP_0} - 1\Big)^2 dP_0 \le K\delta^4.$$

Consequently, let $\delta_n = (Kn)^{-1/4}$ and suppose $\delta_n \bar{b} \le \frac{1}{2}$. For every estimator $\hat{\jmath}$ of $|\delta|$ based on n independent copies of $L(t)$,

$$\max_{\delta \in \{0, \delta_n\}} \mathbb{E}_\delta \big|\hat{\jmath} - |\delta|\big| \ge \frac{\delta_n}{6}.$$

The path on $[0,t]$, in contrast, has Fisher information $t\sum_{i<j} j_{ij}^2/c_{ij} > 0$ about δ at $\delta = 0$.

In words, from n copies of the counts a net flow smaller than about $n^{-1/4}$ cannot be told from no net flow at all, whatever the estimator. The path has positive information about the net flow, so from n copies of the path one expects to go down to about $n^{-1/2}$ (we do not prove this).

Idea of the proof. The likelihood ratio of the path is a product with one factor $1 + \delta b_{ij}$ for each jump from i to j. The counts do not see the jumps, so their likelihood ratio is the average of this product over the paths with the given counts. Run each path backward. Under the reversible chain this does not change the law or the counts, and it turns every factor $1 + \delta b_{ij}$ into $1 - \delta b_{ij}$. So the average only sees the even part of the product, and the even part differs from 1 by order δ^2.

Proof. Step 1: the likelihood ratio of the path. The two chains have the same start and the same exit rates, so the holding times contribute nothing, and each jump from i to j contributes the ratio of the rates. With N_{ij} the number of jumps from i to j up to time t, the likelihood ratio of the path under G_δ against G_0 is

$$R_\delta = \prod_{i \ne j} (1 + \delta b_{ij})^{N_{ij}}.$$

Step 2: run the film backward. The likelihood ratio of the counts is $\frac{dP_\delta}{dP_0}(L) = \mathbb{E}_0[R_\delta \mid L]$. The reversible chain started from π has the same law as its path run backward. Running backward keeps L and swaps N_{ij} with N_{ji}. Since $b_{ji} = -b_{ij}$, it turns R_δ into $R_{-\delta}$. Hence $\mathbb{E}_0[R_\delta \mid L] = \mathbb{E}_0[R_{-\delta} \mid L]$, and

$$\frac{dP_\delta}{dP_0}(L) - 1 = \mathbb{E}_0\Big[\tfrac{1}{2}(R_\delta + R_{-\delta}) - 1 \;\Big|\; L\Big].$$

Step 3: the even part is small. For a given path, R_δ is a product of N factors of the form $1 + \delta b$ with $|b| \le \bar{b}$. So it is a polynomial $\sum_k e_k \delta^k$ with $e_0 = 1$ and $|e_k| \le \binom{N}{k}\bar{b}^k$. The odd powers cancel in $\frac{1}{2}(R_\delta + R_{-\delta})$. With $x = |\delta|\bar{b}$,

$$\Big|\tfrac{1}{2}(R_\delta + R_{-\delta}) - 1\Big| \le \sum_{k \ge 2} \binom{N}{k} x^k = (1+x)^N - 1 - Nx \le \tfrac{1}{2}N^2x^2(1+x)^N,$$

where the last step is Taylor's formula with the second derivative $N(N-1)(1+\xi)^{N-2}$.

Step 4: the chi-square distance. The square of a conditional mean is at most the conditional mean of the square. So by Steps 2 and 3, for $x \le \frac{1}{2}$,

$$\chi^2(P_\delta, P_0) \le \mathbb{E}_0\Big[\big(\tfrac{1}{2}(R_\delta + R_{-\delta}) - 1\big)^2\Big] \le \tfrac{1}{4}\bar{b}^4\delta^4\, \mathbb{E}_0\big[N^4(1+x)^{2N}\big] \le K\delta^4.$$

Here K is finite because the chain jumps at rate at most $\max_i q_i$, so N is at most a Poisson variable with mean $t\max_i q_i$.

Step 5: two hypotheses. For n independent copies the chi-square distance is $(1+\chi^2(P_\delta,P_0))^n - 1 \le e^{nK\delta^4}-1$, which is $e-1$ at $\delta=\delta_n$. The total variation distance between the two laws of the sample is at most half the square root of this, so at most $\frac12\sqrt{e-1}<0.66$. Now use two hypotheses, as in Le Cam's method [23, Chapter 2]. For every value of $\hat\jmath$ we have $|\hat\jmath|+|\hat\jmath-\delta_n|\ge\delta_n$. Integrating this against the smaller of the two laws of the sample, whose total mass is one minus the total variation distance, shows that the two mean errors add up to at least $\delta_n(1-0.66)$. So the larger one is at least $\frac12\delta_n(1-0.66)\ge\delta_n/6$.

The path. By Step 1 the score of the path at $\delta=0$ is $\sum_{i\ne j} b_{ij}N_{ij}$. Its compensator is $\int_0^t \sum_j b_{Y_s j} g^0_{Y_s j}\,ds$, and this vanishes because $\sum_j b_{ij}g^0_{ij} = \sum_j j_{ij}/(2\pi_i) = 0$. So the score is a martingale with variance $\sum_{i\ne j} b_{ij}^2\,\mathbb{E}_0 N_{ij} = \sum_{i\ne j} b_{ij}^2\, t c_{ij}/2 = t\sum_{i<j} j_{ij}^2/c_{ij}$. □

9.1 At a reversible chain the square of the net flow is an ordinary parameter

Theorem 9.1 is a bound from one side. We now show that its power δ^4 is sharp. The counts see δ only through δ^2, and at the reversible chain their likelihood has a first-order expansion in δ^2 with positive information. This gives a test of reversibility within the family G_δ (so π, c and the direction j are known) and the rates of estimation near $\delta=0$.

Two functions of the path are needed. Let

$$W=\sum_{i<j} b_{ij}(N_{ij}-N_{ji}), \qquad V=\sum_{i<j} b_{ij}^2(N_{ij}+N_{ji}), \qquad S=\tfrac12\,\mathbb{E}_0\big[W^2-V \;\big|\; L\big].$$

Here W is the score of the path at $\delta=0$ (the end of the proof of Theorem 9.1). It counts the jumps along the circulation minus the jumps against it, with weights. And V is the variance that W collects along the way, in the sense that W^2-V has mean zero under the reversible chain. So S asks whether the path went around more than chance alone would make it, given the counts.

Theorem 9.2 (the expansion in δ^2)**.** *Let $t>0$ and $|\delta|\bar b\le\frac12$.*

(a) We have $\dfrac{dP_\delta}{dP_0} = 1+\delta^2 S+\rho_\delta$ with $\mathbb{E}_0[\rho_\delta^2]\le K_2^2\,\delta^8$, where $K_2=\frac{1}{24}\bar b^4\big(\mathbb{E}_0[N^8(9/4)^N]\big)^{1/2}$, which is finite for the same reason as K.

(b) The number $I=\mathbb{E}_0[S^2]$ is finite and positive.

(c) The chi-square distance satisfies $\chi^2(P_\delta,P_0)=I\delta^4+O(\delta^6)$.

In words, S is the score of the counts for the parameter δ^2 at $\delta^2=0$, and I is their information about δ^2 at the reversible chain. Part (c) says that the power δ^4 in Theorem 9.1 cannot be improved.

Idea of the proof. The path sees δ to first order through W. Running the film backward flips the sign of W and keeps the counts, so the counts lose this term and keep the next one, which is $\frac12\delta^2(W^2-V)$. The information is positive because of the paths that stay on two states. On a pair $\{i,k\}$ the counts see the weight $g_{ik}g_{ki}$ of a round trip (Theorem 4.1). The two flows of the pair have a fixed sum, so their product is largest when they are equal, and a net flow of either sign makes round trips rarer by a factor $1-\delta^2 b_{ik}^2$.

Proof. (a) By Step 3 of the proof of Theorem 9.1, for a given path $R_\delta=\sum_k e_k\delta^k$, where e_k is the sum of the products of k of the N factors b of the jumps, and $|e_k|\le\binom{N}{k}\bar b^k$. The sum of the factors is W and the sum of their squares is V, so $e_1=W$ and $e_2=\frac12(W^2-V)$. The series is dominated

by $(3/2)^N$, which is integrable, so we may take the conditional mean term by term. Running the film backward keeps the law of the reversible chain and the counts, and it turns each factor b into $-b$, hence e_k into $(-1)^k e_k$. So $\mathbb{E}_0[e_k \mid L] = 0$ for odd k, and

$$\frac{dP_\delta}{dP_0} = \mathbb{E}_0[R_\delta \mid L] = 1 + \delta^2 S + \rho_\delta, \qquad \rho_\delta = \mathbb{E}_0\Big[\sum_{k\ge 4 \text{ even}} e_k \delta^k \;\Big|\; L\Big].$$

With $x = |\delta|\bar{b} \le \frac{1}{2}$, Taylor's formula with the fourth derivative gives $\sum_{k\ge 4} \binom{N}{k} x^k \le \frac{1}{24} N^4 x^4 (1+x)^N$. The square of a conditional mean is at most the conditional mean of the square, so $\mathbb{E}_0[\rho_\delta^2] \le \frac{1}{24^2} \bar{b}^8 \delta^8 \, \mathbb{E}_0[N^8 (9/4)^N]$.

(b) Since $|e_2| \le \frac{1}{2}\bar{b}^2 N^2$, every moment of S is finite. Suppose that $I = 0$, so that $S = 0$ almost surely. Choose a pair $F = \{i, k\}$ with $j_{ik} \ne 0$. (One exists because $\bar{b} > 0$, and then $c_{ik} > 0$.) Let $\mathcal{A}$ be the event that the path stays in F up to time t. Up to a null set this event is a function of the counts (it says that $L_l(t) = 0$ for $l \notin F$). So for every real λ (of either sign, since L is bounded),

$$0 = \mathbb{E}_0\big[S\, e^{-\lambda \cdot L} \mathbf{1}_{\mathcal{A}}\big] = \mathbb{E}_0\big[e_2\, e^{-\lambda \cdot L} \mathbf{1}_{\mathcal{A}}\big]. \tag{6}$$

Put $\Psi(\delta) = \mathbb{E}_\delta[e^{-\lambda \cdot L} \mathbf{1}_{\mathcal{A}}] = \mathbb{E}_0[R_\delta e^{-\lambda \cdot L} \mathbf{1}_{\mathcal{A}}]$. Since $\sum_k |e_k| |\delta|^k \le (3/2)^N$ is integrable, we have $\Psi(\delta) = \sum_k \delta^k \, \mathbb{E}_0[e_k\, e^{-\lambda \cdot L} \mathbf{1}_{\mathcal{A}}]$, so the right side of (6) is the coefficient of δ^2 in Ψ. By the Feynman–Kac formula for the chain killed when it leaves F,

$$\Psi(\delta) = (\pi_i, \pi_k) \, \exp\Big(t \begin{pmatrix} -y_i & g^\delta_{ik} \\ g^\delta_{ki} & -y_k \end{pmatrix} \Big) \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad y_i = \lambda_i + q_i.$$

(The exit rates q_i are the same for every δ.) Take λ with $y_i = y_k = y$. The matrix is then $-y$ plus a matrix whose square is a times the identity, where $a = g^\delta_{ik} g^\delta_{ki} = a^0(1 - \delta^2 b_{ik}^2)$ and $a^0 = c_{ik}^2/(4\pi_i \pi_k) > 0$ is the weight of a round trip in the reversible chain. Since $\pi_i g^\delta_{ik} + \pi_k g^\delta_{ki} = c_{ik}$ does not depend on δ,

$$\Psi(\delta) = e^{-ty} \Big[(\pi_i + \pi_k) \cosh(t\sqrt{a}) + c_{ik} \, \frac{\sinh(t\sqrt{a})}{\sqrt{a}} \Big].$$

Both $\cosh(t\sqrt{a})$ and $\sinh(t\sqrt{a})/\sqrt{a}$ are power series in a with positive coefficients, so the derivative of the bracket in a is positive at $a = a^0$ (here $t > 0$). And Ψ depends on δ only through $a = a^0 - a^0 b_{ik}^2 \delta^2$. Hence the coefficient of δ^2 in Ψ is $-a^0 b_{ik}^2$ times a positive number. It is not zero, which contradicts (6).

(c) By (a), $\chi^2(P_\delta, P_0) = \mathbb{E}_0[(\delta^2 S + \rho_\delta)^2]$, which differs from $I\delta^4$ by at most $2\sqrt{I}\, K_2 |\delta|^6 + K_2^2 \delta^8$. □

Remark 9.3 (the N-step chain). Theorem 9.2 also holds for the chain in discrete time with transition matrix $\mathrm{Id} + \eta G_\delta$, started from π, and at least three visited states (two transitions), as long as $\eta q_i < 1$ for every i, with the vector of visit counts, which we write K as in Section 10, in place of L. Indeed a stay at i has probability $1 - \eta q_i$ for every δ, and a jump from i to j has probability $\eta g^0_{ij}(1 + \delta b_{ij})$. So R_δ is the same product over the jumps, the number of jumps is bounded, and parts (a) and (c) follow as before. For (b) we argue with signs. Take a pair $F = \{i, k\}$ with $j_{ik} \ne 0$, let $\mathcal{A}$ be the event that the path stays in F, and let $z_i, z_k > 0$. A path in F with $2m$ jumps has probability $\pi_x a^m$ times its stay factors, where x is its first state and $a = g^\delta_{ik} g^\delta_{ki} \eta^2$. A path with $2m+1$ jumps that starts at i has probability $\pi_i g^\delta_{ik} \eta \, a^m$ times its stay factors. Run backward, it is

a path that starts at k, with the same counts and the same stay factors, and with $\pi_k g_{ki}^{\delta}$ in place of $\pi_i g_{ik}^{\delta}$. The two add up to $c_{ik}\eta\, a^m$ times the stay factors. Hence

$$\mathbb{E}_\delta\big[z^K \mathbf{1}_{\mathcal{A}}\big] = \sum_{m\ge 0} a^m\, A_m(z),$$

where the A_m do not depend on δ and are sums of nonnegative terms, and $A_1(z) > 0$ because three visited states allow the path i, k, i. The left side depends on δ only through $a = a^0(1-\delta^2 b_{ik}^2)$, where now $a^0 = \eta^2 c_{ik}^2/(4\pi_i\pi_k)$, so its coefficient of δ^2 is $-a^0 b_{ik}^2 \sum_{m\ge 1} m(a^0)^{m-1} A_m(z) < 0$. As in the proof of (b) this coefficient is $\mathbb{E}_0[S\, z^K \mathbf{1}_{\mathcal{A}}]$, so S is not almost surely 0 and $I > 0$. The same argument, with the integral over the jump times in place of the sum over paths, gives another proof of (b) in continuous time.

Now let $L^{(1)}, \dots, L^{(n)}$ be independent copies of the counts, and put

$$T_n = \frac{1}{\sqrt{nI}} \sum_{k=1}^{n} S\big(L^{(k)}\big).$$

Let Z be a standard normal variable and let z_α be the number with $\mathbb{P}(Z > z_\alpha) = \alpha$.

Theorem 9.4 (a test of reversibility)**.** *Let π, c and the direction j be known, so that δ is the only unknown.*

(a) If $\delta = 0$, then $T_n \to Z$ in law.

(b) If $\delta = \delta_n$ with $\sqrt{n}\,\delta_n^2 \to h \ge 0$, then $T_n \to Z + h\sqrt{I}$ in law. So the test that rejects reversibility when $T_n > z_\alpha$ has level tending to α and power tending to $\mathbb{P}(Z > z_\alpha - h\sqrt{I})$.

(c) If $\sqrt{n}\,\delta_n^2 \to 0$, then every test with level at most α has power at most $\alpha + o(1)$ against δ_n.

In words, the test is one-sided, because a net flow of either sign pushes T_n up. It sees a net flow of size $n^{-1/4}$, and by (c) no test sees a smaller one, even when everything but δ is known.

Proof. (a) Under P_0 the variable S has mean $\mathbb{E}_0[\mathbb{E}_0[e_2 \mid L]] = \frac{1}{2}\mathbb{E}_0[W^2 - V] = 0$ (the variance of W is $\mathbb{E}_0 V$, as at the end of the proof of Theorem 9.1) and variance I. So this is the central limit theorem.

(b) Since $\delta_n \to 0$ we have $|\delta_n|\bar{b} \le \frac{1}{2}$ for large n. Write $\ell_\delta = dP_\delta/dP_0$. By Theorem 9.2(a) and the Cauchy–Schwarz inequality,

$$\mathbb{E}_\delta S = \mathbb{E}_0[S\ell_\delta] = \delta^2 I + \mathbb{E}_0[S\rho_\delta], \qquad |\mathbb{E}_0[S\rho_\delta]| \le \sqrt{I}\, K_2\, \delta^4,$$

$$\big|\mathbb{E}_\delta S^2 - I\big| = \big|\mathbb{E}_0[S^2(\ell_\delta - 1)]\big| \le \big(\mathbb{E}_0 S^4\big)^{1/2}\big(\sqrt{I}\,\delta^2 + K_2\delta^4\big),$$

and $\mathbb{E}_\delta|S|^3 \le (\mathbb{E}_0 S^6)^{1/2}(1 + \sqrt{I}\delta^2 + K_2\delta^4)$. So along $\delta_n \to 0$ the mean of S is $\delta_n^2 I + O(\delta_n^4)$, its variance $\mathbb{E}_\delta S^2 - (\mathbb{E}_\delta S)^2$ tends to I, and its third absolute moment stays bounded (and with it the third absolute moment about the mean, which is at most eight times as large). By Lyapunov's central limit theorem the sum $\sum_k S(L^{(k)})$, centered and divided by $\sqrt{nI}$, tends to Z in law. The centering is $\sqrt{n}\,(\delta_n^2 I + O(\delta_n^4))/\sqrt{I} \to h\sqrt{I}$.

(c) The bound in Step 5 of the proof of Theorem 9.1 holds for every δ with $|\delta|\bar{b} \le \frac{1}{2}$. So for large n the total variation distance between the laws of the sample under δ_n and under 0 is at most $\frac{1}{2}(e^{nK\delta_n^4} - 1)^{1/2}$, which tends to 0. The power of a test exceeds its level by at most this distance. □

Corollary 9.5 (the two rates). *Let* $\hat{s} = \max\bigl(0, \frac{1}{nI}\sum_k S(L^{(k)})\bigr)$*, and estimate the size of the net flow by* $\sqrt{\hat{s}}$ *and the entropy production by* $\hat{\sigma} = 2\hat{s}\sum_{i<j} j_{ij}^2/c_{ij}$*. There is a constant* C*, depending on* π*,* c*,* j *and* t *only, such that, for every* n *and every* δ *with* $|\delta|\bar{b} \le \frac{1}{2}$*,*

$$\mathbb{E}_\delta\bigl|\hat{s} - \delta^2\bigr| \le C\bigl(n^{-1/2} + \delta^4\bigr), \qquad \mathbb{E}_\delta\bigl|\hat{\sigma} - \sigma(\pi, G_\delta)\bigr| \le C\bigl(n^{-1/2} + \delta^4\bigr),$$

$$\mathbb{E}_\delta\bigl|\sqrt{\hat{s}} - |\delta|\bigr| \le \min\Bigl(\sqrt{C}\,\bigl(n^{-1/4} + \delta^2\bigr),\ \frac{C}{|\delta|}\bigl(n^{-1/2} + \delta^4\bigr)\Bigr).$$

In words, as long as $|\delta|$ is at most of order $n^{-1/8}$, the entropy production is learned to within $n^{-1/2}$. In this range the entropy production itself can be as large as $n^{-1/4}$, so this says something. The size of the net flow is learned to within $n^{-1/4}$ at $\delta = 0$, which is the rate of Theorem 9.1, and to within $n^{-1/2}/|\delta|$ for larger δ in this range. (Outside this range $\hat{s}$ is biased, and one would use maximum likelihood.) At $\delta = 0$, $\sqrt{n}\,\hat{s}$ tends in law to the positive part of a normal variable with variance $1/I$, by Theorem 9.4(a), and so does $\sqrt{n}\,\hat{\sigma}$ up to a constant factor. So intervals based on a normal limit fail at a reversible chain.

Proof. Let m be the mean of the $S(L^{(k)})$. By the bounds in the proof of Theorem 9.4(b), which hold for $|\delta|\bar{b} \le \frac{1}{2}$, the mean of m/I differs from δ^2 by at most $K_2\delta^4/\sqrt{I}$, and its variance is at most a constant over n. So $\mathbb{E}_\delta|m/I - \delta^2| \le C(n^{-1/2} + \delta^4)$. Taking the positive part moves m/I toward $\delta^2 \ge 0$, so the same bound holds for $\hat{s}$. For the entropy production, Remark 6.2(2) gives $\sigma(\pi, G_\delta) = \sum c_{ij}\,\varphi(\delta^2 b_{ij}^2) = 2\delta^2\sum j_{ij}^2/c_{ij} + O(\delta^4)$, where the sums are over the pairs with $c_{ij} > 0$ and the error is uniform because $\delta^2 b_{ij}^2 \le \frac{1}{4}$. For the size of the net flow use $|\sqrt{u} - \sqrt{v}| \le \sqrt{|u - v|}$ and Jensen's inequality for the first bound, and $|\sqrt{u} - \sqrt{v}| \le |u - v|/\sqrt{v}$ for the second. □

Remark 9.6. (1) To leading order $\sigma = 2\delta^2\sum_{i<j} j_{ij}^2/c_{ij}$, and $\sum_{i<j} j_{ij}^2/c_{ij}$ is the information of the path about δ per unit time (Theorem 9.1). So near a reversible chain the entropy production is twice the square of the net flow, measured in the units in which the path sees it.

(2) The estimator $\hat{s}$ is built for a neighborhood of $\delta = 0$. Away from it one would use maximum likelihood. For a regular model with the true parameter on the boundary, the standard theory [7, 22] says that at $\delta = 0$, $\sqrt{n}$ times the maximum likelihood estimate of δ^2 tends to the positive part of a normal variable with variance $1/I$, and that twice the logarithm of the likelihood ratio tends to a variable that is 0 with probability $\frac{1}{2}$ and otherwise is the square of a standard normal variable. Theorem 9.2 supplies the expansion and the positive information that this theory starts from. We have not checked its remaining conditions (for one, we have not shown that the model is regular at $\delta \neq 0$). The simulations of Remark 9.7, which are for an N-step chain, agree with it.

(3) We took π, c and the direction j as known. For three states there is only one direction. With π and c unknown one expects the same rates, since those are learned at the rate $n^{-1/2}$. For this the score S must not lie in the span of the scores of the counts for π and c (if it did, a change of π and c could imitate a small net flow). We have proved neither this nor the rates. For four or more states with every $c_{ij} > 0$ the circulations form a space of dimension $\binom{d-1}{2}$. The law of the counts is even in the circulation, so to leading order it can depend on it only through a quadratic form. We have not worked this case out.

Remark 9.7 (the rates are attained in an example). Take three states, the N-step chain with $N = 40$ and $T = 2$, $\pi = (0.5, 0.3, 0.2)$ and two-way flows 0.30, 0.20, 0.25, all known, and let the only unknown be the flow j around the triangle. The law of the count vector is even in j (the proof of Theorem 9.1 works step by step for this chain), and its chi-square distance from the law at

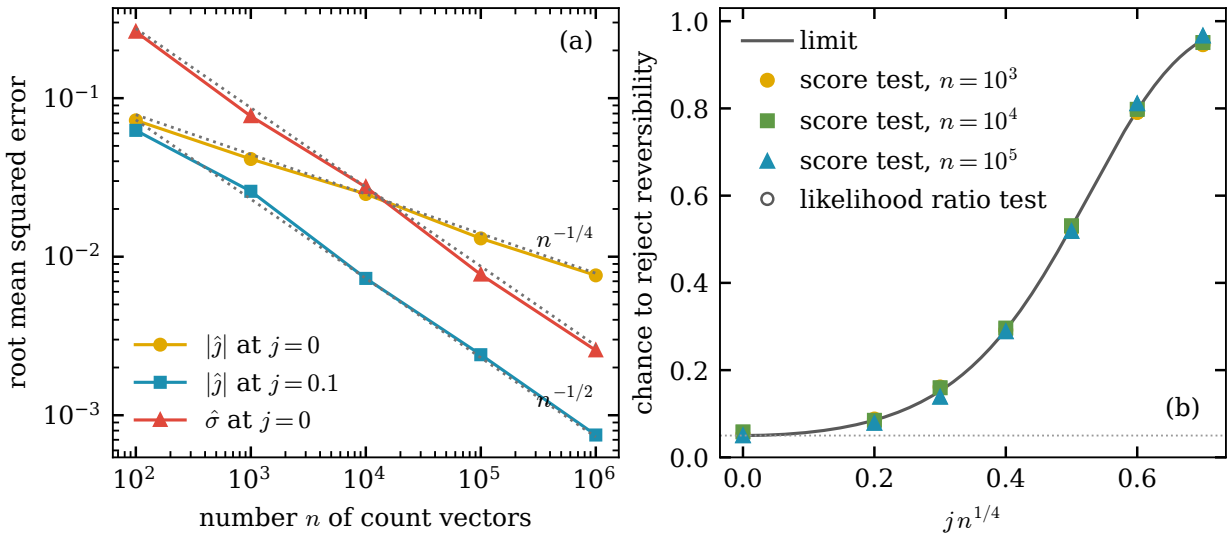


Figure 3: The example of Remark 9.7. (a) Root mean squared errors from 400 samples of n count vectors each. At a reversible chain the net flow is learned at the rate $n^{-1/4}$ and the entropy production at the rate $n^{-1/2}$. Away from reversibility the net flow is learned at the rate $n^{-1/2}$. The dotted lines have these slopes. (b) The chance that a test of level 5% rejects reversibility, against $j\,n^{1/4}$, from 2000 samples for each point. Filled marks are the score test of Theorem 9.4 and open marks the likelihood ratio test. The curve is the limit $\mathbb{P}(Z > z_{0.05} - \sqrt{I}\,j^2\sqrt{n})$ with $I = 47.1$. The three sample sizes fall on one curve, so the test sees a net flow of size $n^{-1/4}$.

$j = 0$ is $47.1\,j^4$ for small j. We estimate $|j|$ by maximum likelihood from n count vectors, and the entropy production by plugging in (Figure 3). At $j = 0$ the error of $|\hat{\jmath}|$ falls like $n^{-0.248}$, and the error of $\hat{\sigma}$ like $n^{-0.498}$. At $j = 0.1$ the error of $|\hat{\jmath}|$ falls like $n^{-0.509}$ (slopes fitted for $n \geq 10^3$). So the rate $n^{-1/4}$ of Theorem 9.1 is attained here, and the entropy production is learned at the usual rate even where the net flow is not, as Remark 6.2 suggests. The numbers fit the standard theory of a parameter on the boundary [22]. In the coordinate $s = j^2$ the model is smooth with information $I = 47.1$ at $s = 0$, the maximum likelihood estimate $\hat{\jmath}^2$ of s is close to the positive part of a normal variable with variance $1/(nI)$, and $|\hat{\jmath}|$ then has root mean squared error $(2\pi)^{-1/4}(nI)^{-1/4}$. At $n = 10^6$ this is 0.00762, and the simulation gives 0.00762. (A root mean squared error from 400 samples has a relative error of about 4%, so the agreement to three digits is partly luck.) The score S of Theorem 9.2, computed for this chain, has $\mathbb{E}_0 S^2 = 47.1$, the same number, as part (c) of that theorem says (Remark 9.3). We also ran the test of Theorem 9.4 and the likelihood ratio test at level 5% (Figure 3(b)). The likelihood ratio test rejects when twice the logarithm of the likelihood ratio exceeds $z_{0.05}^2 = 2.71$, the critical value of the limit law in Remark 9.6(2). At $j = 0$ they reject in 4.9% to 5.8% of 2000 samples, and the likelihood ratio is exactly 1 in 51% of them. For $j\,n^{1/4} = 0.5$ the limit of the power is 0.528, and the score test rejects in 52.4%, 53.1% and 51.8% of the samples for $n = 10^3$, 10^4 and 10^5. (A share from 2000 samples has a standard error of at most 1.1 percentage points. Theorem 9.4 is stated for continuous time. Its proof uses Theorem 9.2 and the chi-square bound of Theorem 9.1, and both hold for this chain (Remark 9.3).)

10 Discrete time, the first $2d$ horizons

Papers A to C count the steps of a chain in discrete time. The proof of Theorem 3.1 does not carry over (powers of algebraic functions can cancel where exponentials cannot). But if we are given the counts at the horizons 1 to $2d$, the algebra of Sections 4 to 6 does. (We have not carried over the lumping, the information or the rates.)

Let P be an irreducible transition matrix on $[d]$, let μ be a start law, and let K_N be the vector of visit counts of the first N steps. A cut of P is defined as for G, and the reversal is $\hat{p}_{ij} = \pi_j p_{ji}/\pi_i$.

Proposition 10.1 (discrete time)**.** *The laws of K_N for $1 \le N \le 2d$ determine every principal minor of P and every principal minor of $P - \mathbf{1}\mu^\top$, and these minors determine the law of K_N for every $N \ge 1$. If P has no cut and (μ', P') gives the same laws for $1 \le N \le 2d$, then $\mu' = \mu$, and P' is P or its reversal.*

In words, with the first $2d$ horizons in hand the discrete chain is read in the same way as the continuous one, and later horizons add nothing. Reading the minors of order one and two as in Section 4, the counts fix μ_i, p_{ii}, $p_{ij}p_{ji}$ and $\mu_i p_{ij} + \mu_j p_{ji}$. If every μ_i is positive, this gives the unordered pairs of flows $\{\mu_i p_{ij}, \mu_j p_{ji}\}$, and with them (when $p_{ij} > 0$ exactly when $p_{ji} > 0$) the entropy production $\sum_{i<j} (\mu_i p_{ij} - \mu_j p_{ji}) \log \frac{\mu_i p_{ij}}{\mu_j p_{ji}}$ of the discrete chain.

Proof. We collect all horizons in one generating function. A path $x_1, \dots, x_N$ contributes $\mu_{x_1} p_{x_1 x_2} \cdots p_{x_{N-1} x_N}$ times $z_{x_1} \cdots z_{x_N}$ to $\mathbb{E} z^{K_N}$, so $\mathbb{E} z^{K_N} = \mu^\top D_z (P D_z)^{N-1} \mathbf{1}$ with $D_z = \operatorname{diag}(z)$. Multiply by s^N, sum over N and put $y_i = s z_i$. For small y,

$$\sum_{N \ge 1} s^N \, \mathbb{E} z^{K_N} = \mu^\top D_y (I - P D_y)^{-1} \mathbf{1} = \frac{n(y)}{\chi(y)}, \qquad \chi(y) = \det(I - P D_y),$$

and $n(y) = \det(I - P D_y + \mathbf{1}\mu^\top D_y) - \chi(y)$ by the formula for the determinant of a rank one change. For a matrix A we have $\det(I - A D_y) = \sum_U (-1)^{|U|} \det(A_{UU}) \prod_{i \in U} y_i$. So the coefficients of χ are the principal minors of P, and those of $\chi + n$ are the principal minors of $P - \mathbf{1}\mu^\top$.

It remains to see that the fraction is in lowest terms. Both χ and n have degree at most one in each y_i, and $\chi(0) = 1$ while $n(0) = 0$. With $x_i = 1/y_i$ we have $\chi(y) \prod_i x_i = \det(X - P)$, which is irreducible by the proof of Lemma 3.3. (Put $\tilde{x}_i = x_i - p_{ii}$. Then $\det(X - P)$ is the polynomial of that proof with A the part of P off the diagonal.) So χ is irreducible. (If χ were a product of two nonconstant factors, each would have degree at most one in each variable and a nonzero constant term, the two would use different variables, and replacing y_i by $1/x_i$ in each and clearing denominators would factor $\det(X - P)$.) Also χ has degree exactly one in each y_i, since otherwise x_i would divide $\det(X - P)$. So if χ divided n, then n would be a constant multiple of χ, and the constant would be $n(0)/\chi(0) = 0$. But the coefficient of y_i in n is μ_i. Hence the fraction is in lowest terms. Now let a second pair (μ', P') give the same laws for $1 \le N \le 2d$. The term with s^N is homogeneous of degree N in y, so the power series of $n/\chi - n'/\chi'$ has no term of degree at most $2d$. Since $\chi\chi'$ has constant term 1, the same holds for the polynomial $n\chi' - n'\chi$. But n, n', χ and χ' have degree at most d, so this polynomial has degree at most $2d$, and it is 0. So $n'\chi = n\chi'$, hence χ divides χ', and then $\chi' = \chi$ because χ' also has degree at most one in each variable and constant term 1. So the first $2d$ laws determine χ and n, and χ and n determine every law. This is the first claim. The coefficient of y_i also gives $\mu' = \mu$.

For the second claim, the theorem quoted before Theorem 5.2 gives $P' = DPD^{-1}$ or $P' = DP^\top D^{-1}$. The rows of P' sum to one. In the first case $PD^{-1}\mathbf{1} = D^{-1}\mathbf{1}$, so $D^{-1}\mathbf{1}$ is a multiple of $\mathbf{1}$ and $P' = P$. In the second case $P^\top D^{-1}\mathbf{1} = D^{-1}\mathbf{1}$, so $D^{-1}\mathbf{1}$ is a multiple of π and $P' = \hat{P}$. □

11 An example with data, the time budgets of sleep

We try the theory on data where the full path is known, so that we can throw it away and see what the time budgets alone give back.

The data. The Sleep-EDF telemetry study [14, 20] has 44 nights of sleep, two each from 22 subjects, scored in epochs of 30 seconds. (The subjects had mild difficulty falling asleep, and one of each subject's two nights followed a dose of temazepam and the other a placebo. We pool all nights.) We use three states, namely awake, non-REM sleep (stages 1 to 4) and REM sleep. Epochs scored as movement time count as awake. We cut off the waking time before the first and after the last epoch of sleep, cut each night into windows of 60 epochs (30 minutes), and drop what is left at the end of the night (about 3% of the epochs). This gives 659 windows (Figure 4(a)). For each window the *time budget* is the number of epochs in each state, and it is all we keep. Time budgets are how sleep and activity data are usually summarized, and they are compositional data [1, 6] (Figure 4(b)).

Two fits. The data are in discrete time, so we use the chain of Section 10 with a stationary start, in the coordinates (π, c, j), where j is the flow around the triangle awake, non-REM, REM. The first fit uses the budgets alone, by maximum likelihood with the exact law of the count vector, treating the windows as independent copies. The law of the budgets is even in j (run the film backward), so the fit can only give $|j|$. This fit is a stationary chain. The second fit uses the full sequences of the same windows, by counting the transitions inside the windows. It is not constrained to be stationary. Its flows are the transition counts per epoch, so its net flows on the three pairs need not be equal. Table 2 and Figure 4(c), (d) give the results. The intervals run from the 5th to the 95th percentile of 60 resamples of the nights, so they are rough.

	full sequences	budgets only
share of time (awake, non-REM, REM)	0.060, 0.735, 0.205	0.061, 0.744, 0.195
two-way flow awake, non-REM	32.8 (28.8, 37.2)	18.4 (15.8, 21.5)
two-way flow awake, REM	3.3 (2.7, 3.7)	2.9 (2.3, 3.6)
two-way flow non-REM, REM	15.8 (13.3, 18.9)	8.8 (7.6, 9.9)
net flow around the triangle	2.3 (1.9, 2.6)	$\lvert j\rvert = 2.9$ (2.1, 3.6), on the boundary

Table 2: Sleep as a chain on three states, fitted to the full sequences and to the time budgets alone. Flows are per 1000 epochs, with 90% intervals from resampling the nights. In the full sequences the net flow runs from awake to non-REM to REM and back to awake (the three pairs give 2.2, 2.4 and 2.2, and we show their mean). In the fit from the budgets the net flow equals the two-way flow of the pair awake, REM, which is the largest value the model allows.

What we see. (1) The fit from the budgets gives the share of time in each state to within 0.01.

(2) The budgets say little about the net flow. The full sequences show a clear one. Sleep goes around the triangle awake, non-REM, REM, awake about 2.3 times in 1000 epochs more often than the other way. The fit from the budgets gives $|j| = 2.9$, which looks close. But this value is not an interior maximum of the likelihood. The likelihood keeps rising in $|j|$ until the net flow equals the two-way flow of the pair awake, REM, and the estimate is this bound. Against the best fit with no net flow the likelihood ratio statistic is 5.3. If the windows were independent and sleep were a chain, this would exceed the 5% critical value 2.71 of Remark 9.7. Neither holds, so

we attach no level to it. The budgets are compatible with a net flow of this size, and that is all. The same happens on data simulated from the chain fitted to the full sequences, with the same number of windows. In 24 runs the estimate of $|j|$ was close to 0 in 11 of them, on the bound in 8, and had mean 1.6 and standard deviation 1.7, where the true value is 2.3. This is in line with Theorem 9.1, although that theorem is for continuous time, for a chain close to reversible, and with the rest of the chain known (and sleep is not close to reversible, since on the pair awake, REM the net flow is two thirds of the two-way flow). A net flow is hard to see in the counts, and 659 windows are too few.

(3) The budgets give two-way flows that are too small, by almost half on the two large pairs. This is not a fault of the method. In the 24 simulated runs the budgets gave 32.7, 3.4 and 16.2 on average for the three two-way flows (standard deviations 1.6, 0.6 and 1.2), and the true values are 32.7, 3.3 and 15.8. We think the cause is that sleep is not a chain with fixed rates, and we have not tested this. The data show two ways in which the chain fails. First, a chain forces the length of a waking bout to be geometric, and the data are far from it (Figure 4(e)). Half of the waking bouts in these nights last a single epoch, where the chain predicts 30%, and 2.1% last ten minutes or more, where the chain predicts 0.1%. Heavy tails of brief awakenings are well known [16]. A plausible reason for the small flows is that many short awakenings cost little time, so the budgets hardly see them. Second, sleep changes through the night (deep sleep comes early and REM sleep late), so windows from different hours do not have one law. This spreads the budgets more than a single chain would, and a chain fitted to them may answer by switching less often. The first effect is not confined to waking. Non-REM bouts are not geometric either (19% of them last a single epoch, where the chain predicts 3%), so it also reaches the pair non-REM, REM, which is halved as well. We have not separated the two effects.

(4) We do not trust the entropy production here. The full sequences give $\sigma = 4.7$ per 1000 epochs, with interval $(3.4, 6.1)$. In the fit from the budgets the pair awake, REM is crossed in one direction only, and there the formula of Theorem 6.1 blows up.

So on real data the example supports less than we hoped. The share of time is recovered. In 24 simulations from one fitted chain the two-way flows are recovered, and for sleep they are not. The net flow is barely seen at this sample size, with or without a true chain.

Limits. The windows of one night are not independent, and the first window of each night starts with sleep by construction. The two nights of a subject are not independent either, and we resample nights, not subjects. The nights with and without the drug are pooled. The window of 30 minutes is a choice, and we tried no other. Merging stages 1 to 4 hides the structure inside non-REM sleep. Proposition 10.1 uses the horizons 1 to $2d$, and we use one. Since the model does not hold, what the fit from the budgets estimates has no fixed relation to the flows of the full sequences.

12 Checks and open questions

Most statements above were checked numerically in the scripts next to this note. (Theorem 3.1 and the bound of Theorem 9.1 were not, apart from the growth like δ^4 in one example.)

- `check_identifiability.py`. Reversal agrees to 10^{-15} for $d = 3, 4, 5$. The map from G to the law has full rank at random points for $d = 2, 3, 4$ with a known start, a stationary start and a start at one state. At reversible chains with a stationary start the rank drops by

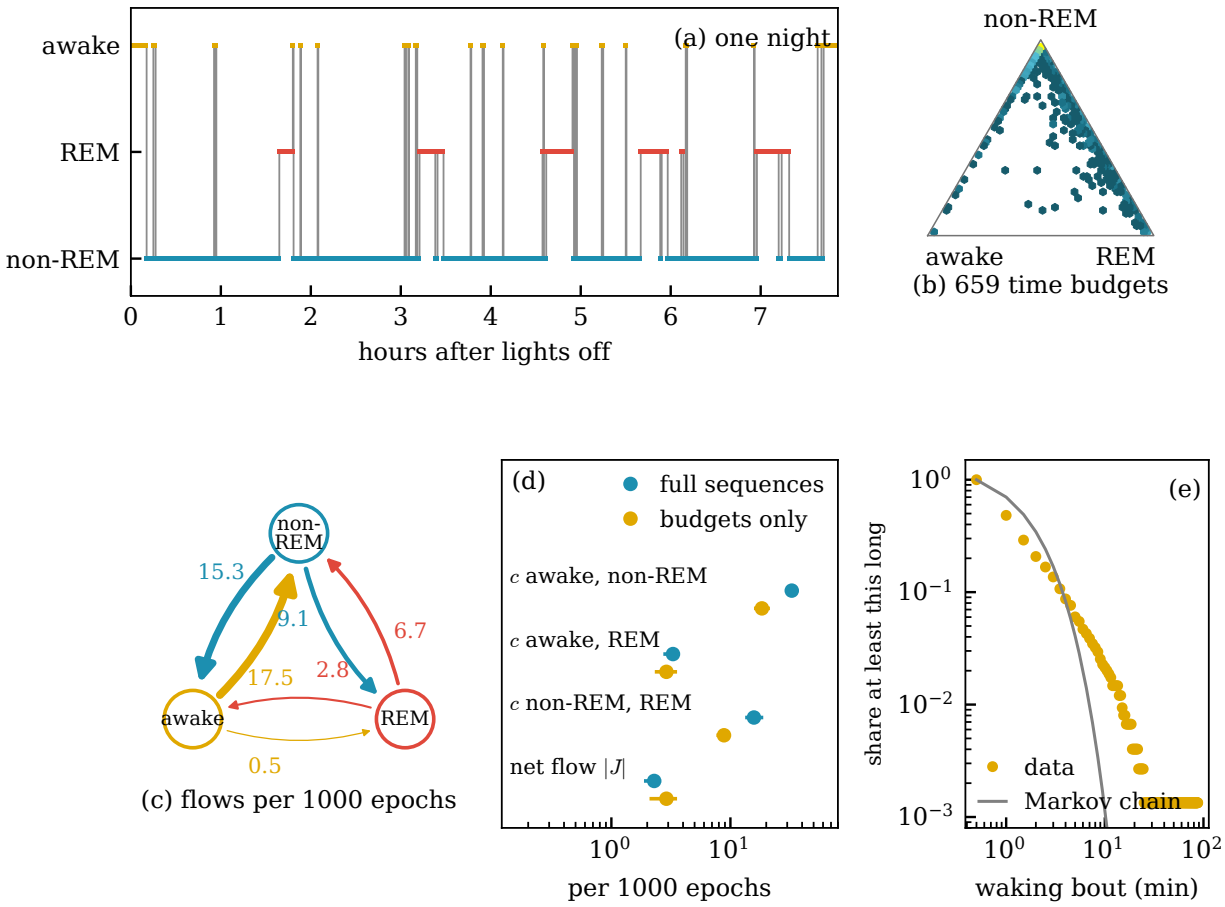


Figure 4: Sleep and its time budgets. (a) One night, as a hypnogram. (b) The 659 time budgets on the simplex, on a logarithmic color scale. Most windows are spent in non-REM sleep, with some time awake or in REM. (c) The flows of the chain fitted to the full sequences, per 1000 epochs. (d) The two-way flows and the net flow from the full sequences and from the budgets alone, with 90% intervals. (e) The share of waking bouts that last at least a given time, against the geometric law of the fitted chain.

exactly $\binom{d-1}{2}$ for $d = 3, 4, 5$. Block reversal at a cut state works when the return rates are equal and fails (by $2 \cdot 10^{-3}$) when they are not.

- `check_lumping_decoder.py`. Example 7.3. The recipe of the remark after Corollary 4.3 returns one generator for a random start and two for a stationary start, for $d = 3, 4, 5$. For the N-step chain on three states the map from P to the law of K_N has full rank once $N \geq 3$.
- `check_information.py`. The two leading terms of Theorem 8.1 (the first to within 0.017 at $T = 0.025$, the second within 3% at $T = 0.02$ and still converging), Theorem 8.2 and Table 1.
- `check_entropy.py`. Theorem 6.1. The entropy production is recovered from the function r alone, to four digits, for $d = 3, 4, 5$ and both kinds of start. Chains with the same counts have the same σ, and the triangle affinities are recovered from the principal minors of orders up to three.
- `sleep_data.py`, `sleep_application.py`, `sleep_checks.py` (the likelihood ratio and the 24

simulated runs) and `sleep_figure.py`. Section 11. The hypnogram files are in the folder `data_sleep_edf` with their checksums.

- `sim_rates.py`. Remark 9.7 and Figure 3. The chi-square distance grows by the factors 15.97 and 15.89 when δ doubles. The score of Theorem 9.2 has $\mathbb{E}_0 S^2 = 47.09$, and the chi-square distance divided by δ^4 is 47.07 at $\delta = 0.02$. The mean of the score at $j = 0.02$ and 0.05 is 1.000 and 0.998 times Ij^2. The two tests of reversibility are run on 2000 samples for each of three sample sizes and seven sizes of the net flow.
- `check_cut_transpose.py`. The lumped reversal of Example 7.3 is the transpose across the module. In three random examples, a transpose across a cut that is not a module keeps the principal minors of G but not the counts. The generating function of Proposition 10.1 and its two expansions in principal minors, for $d = 3, 4$.
- `check_rigidity.py`. Part (b) of Theorem 7.1 and Example 7.2 (the laws agree to 10^{-15} for $d = 4, 5, 7$). In each tie both sets of minors of Proposition 5.1 agree. For a random chain with a start that is not stationary, the reversal has the same minors of G but not of $G - \mathbf{1}\mu^\top$, and the laws differ.

What we do not know.

1. Is every tie built from the two reversals of Theorem 7.1? For a chain with no cut the answer is in Theorem 5.2, up to deciding (4). For a chain with a cut, the results of [5] list every G' with the same principal minors as G, so what is left is to find which of them also pass the test with $G - \mathbf{1}\mu^\top$. For four states with every rate positive, one step across one cut is enough to reach every such G' [5, Lemma 3.1].
2. When does (4) hold? A stationary start and Example 7.2 are the cases we know.
3. Does Theorem 3.1 have a form for the N-step chain? The proof uses that exponentials of algebraic functions do not cancel, and powers of algebraic functions can. Proposition 10.1 uses the horizons 1 to $2d$. Is one horizon $N \geq 3$ enough? (The rank check in `check_lumping_decoder.py` suggests that it is for three states.)
4. The large T row of Table 1. By Theorem 8.4, $d - 1$ efficiencies tend to 1. We expect the rest to tend to 0. For the chain of Table 1 three of them fall off like $1/T$ and the smallest like T^{-2}, and we do not know the general pattern.
5. A start at one state. The rank check says almost every chain is recovered, and the pair formulas change because some μ_i vanish. (Proposition 5.1 and Theorem 5.2 do not need $\mu_i > 0$.)

What is in the literature. A search (not a full review) found the following. The closest work in probability is by Kim and Landim [15], who ask when the large deviation rate function of the occupation times (the limit of our data as the horizon grows) determines the chain. They show that it determines the exit rates, the products $g_{ij}g_{ji}$ and the stationary law, and with them the generator of a reversible chain, and they give a cycle on three states that it cannot tell from its reversal (their Section 5). Our data are the law at one finite horizon, with any start. This law holds more. It sees the start, so away from a stationary start it determines almost every chain (Corollary 4.3), it gives the unordered pairs of flows and the entropy production (Theorems 4.1 and 6.1), and for it we can say exactly which chains tie when there is no cut. The algebra of Section 5 and of the polynomial χ_0 in Section 7 is the theory of matrices with equal principal

minors [11, 17, 3, 5]. We did not find Theorem 3.1, the role of the start law (the unordered pairs of flows in Theorem 4.1, the second matrix in Proposition 5.1, condition (4), and Theorem 7.1 as a statement about the law of $L(t)$), or Theorems 6.1, 8.1 and 8.2. Work on aggregate data from Markov chains and on the embedding problem observes something else, namely how many of a population sit in each state at given times. Work on aggregated Markov processes (see [9]) observes a lumped path over time, with its order, and not the total times. We have not read [11] and [17] themselves. Their statements are taken from [3] and [5], which agree.

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